

Master's Thesis

Predictive Multiplicity in Rashomon Sets via Observation-Wise Variance: Identifying Subgroups by Features and Hyperparameter Sensitivity

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Abstract

Modern machine learning pipelines often produce many models with near-identical predictive performance, yet these models can disagree substantially on individual predictions—a phenomenon known as *predictive multiplicity*. Existing metrics quantify this disagreement globally but do not reveal *where* in the feature space instability concentrates or *what* drives it.

This thesis proposes a spatial analysis framework for predictive multiplicity. We compute observation-wise predictive variance across a Rashomon set of near-optimal models and apply spatial statistics—global Moran’s I and Local Indicators of Spatial Association (LISA)—to detect, localize, and test the significance of multiplicity hotspots in feature space. We decompose variance into between-family and within-family (hyperparameter) components and extract interpretable rules that describe high-multiplicity regions.

Experiments on three benchmark datasets (COMPAS, German Credit, Breast Cancer) with five model families, 250 candidate models, and 10 independent runs each show that: (i) predictive variance is spatially clustered (Moran’s I significant in 90% of runs); (ii) model family explains 41–62% of prediction variance, rising to 52–69% inside hotspots; (iii) hotspot regions are robust to the Rashomon set size, neighborhood graph density, post-hoc calibration, and alternative distance metrics; and (iv) on COMPAS, hotspots are not reducible to decision-boundary proximity and can be described by compact rules involving age and prior criminal history. Validation on synthetic data with known ambiguous regions confirms that the pipeline recovers ground-truth instability with precision 0.99 and recall 0.91.

These findings demonstrate that predictive multiplicity is not merely a global nuisance but a spatially structured phenomenon that concentrates in identifiable subpopulations, with implications for the reliability, fairness, and interpretability of automated decision systems.

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1 Introduction

Supervised machine learning (ML) models are widely used to support decision making in areas such as healthcare, finance, and criminal justice. Their appeal lies in their strong predictive performance, but this often comes at the cost of transparency: multiple models with similar accuracy may behave differently, making their predictions difficult to interpret and trust. Classical evaluation metrics such as accuracy or AUC provide only a coarse summary of model quality and offer little insight into the stability of individual predictions.

Recent work shows that modern ML pipelines typically yield not a single model, but a collection of distinct models that perform almost equally well on the same task. These models may disagree in their predictions for specific individuals, despite being indistinguishable in terms of overall predictive accuracy. This raises fundamental questions about the reliability, fairness, and interpretability of automated decision systems: if several equally accurate models exist, which one should be trusted, and how stable are the conclusions drawn from any single one?

In this work, we study this phenomenon—known as *predictive multiplicity*—by examining how predictions vary across sets of near-optimal models, whether this variation exhibits spatial structure in feature space, and what drives it. We propose a spatial analysis framework that combines observation-wise predictive variance with tools from spatial statistics to identify, localize, and interpret regions of instability.

1.1 Motivation

Automated decision systems are increasingly used to support consequential decisions in domains such as criminal justice, credit approval, healthcare, and social services. These systems typically rely on predictive models trained on historical data, and the selected model is often treated as if it were the only, definitive representation of the relationship between inputs and outcomes. In practice, however, modern machine learning pipelines often produce many different models that achieve nearly identical accuracy. These models all appear equally valid from a performance perspective, yet they can still produce different predictions for the same individual. The following fictional examples illustrate how the existence of multiple, equally accurate models can lead to instability, and why this undermines the reliability of automated decision systems.

1.1.1 Example 1: Prediction Instability at the Individual Level

Jordan is evaluated by an automated risk assessment system. The deployed model predicts a risk of reoffending within two years with a probability of 0.29, classifying Jordan as “low risk.” During an audit, the defense gains access to all models that perform within a small tolerance of the optimal validation accuracy. Although these models are equally accurate overall, they produce substantially different predictions for Jordan, ranging from 0.22 to 0.54.

The prediction-instability score, defined as the variance of Jordan’s predicted risk across all equally accurate models, is 0.017, placing Jordan among the most unstable individuals

in the dataset. Whether Jordan is assessed as low-risk or high-risk depends not on the data alone, but on which near-optimal model the training procedure happened to select.

1.1.2 Example 2: Spatial Clustering of Instability

An auditor discovers that individuals like Jordan are not isolated cases. By mapping prediction instability scores onto the feature space of all individuals evaluated by the system, a clear pattern emerges: high-instability individuals cluster together in specific regions of the input space. Using a spatial autocorrelation statistic, the auditor confirms that this clustering is statistically significant and not due to chance.

The auditor identifies contiguous groups of individuals whose predictions are systematically unstable across near-optimal models. These regions correspond to subpopulations for whom the system’s output is particularly unreliable: the specific model chosen during training determines their classification outcome more than their actual characteristics do.

1.1.3 Example 3: Drivers of Instability

To understand *why* certain individuals land in instability hotspots, the auditor decomposes the prediction variance. About 41–62% of the total prediction variance turns out to be explained by model family (e.g., neural networks vs. tree-based models vs. logistic regression), rising to 52–69% inside hotspots. Within each family, specific hyperparameters drive additional variance: for neural networks, the learning rate and architecture matter most; for tree-based models, the maximum tree depth dominates.

Moreover, the drivers differ between hotspot and non-hotspot regions. Inside the instability hotspots, neural network architecture choices explain more variance than they do globally, suggesting that these regions are where architectural decisions have the most consequential impact on predictions.

Together, these examples reveal that predictive multiplicity is not merely a global nuisance but a spatially structured phenomenon: it concentrates in identifiable regions of the feature space, affects specific subpopulations, and is driven by identifiable modeling choices. Understanding where such instability occurs and what drives it is essential for the development of reliable and transparent automated decision systems.

1.2 Predictive Multiplicity and Rashomon Sets

The phenomenon illustrated above has been formalized under two related concepts. *Predictive multiplicity* (Marx et al., 2020) refers to the situation where models with similar overall performance produce conflicting predictions for individual observations. The *Rashomon effect* (Breiman, 2001) describes the broader observation that many distinct models can explain the same data equally well.

The set of near-optimal models is called the *Rashomon set*:

$$\mathcal{R}_\varepsilon = \{f \in \mathcal{F} : L(f) \leq L^* + \varepsilon\},$$

where L^* is the best achievable loss and ε controls the tolerance. Prior work has studied predictive multiplicity through metrics such as ambiguity (the range of predictions),

discrepancy (maximum pairwise disagreement), and disagreement rate (fraction of observations where models disagree beyond a threshold). In this thesis, ambiguity and discrepancy follow the probability-space definitions used in the *Resolving Predictive Multiplicity for the Rashomon Set* framework. However, these metrics treat observations independently and do not capture spatial structure—whether instability concentrates in specific regions of the feature space, how stable these regions are, or what drives them.

1.3 Problem Statement

Predictive models with similar accuracy can differ substantially in their predictions. Such variation arises naturally in modern machine learning pipelines due to changes in model family, hyperparameters, random seeds, or optimization paths. Standard evaluation metrics do not reveal whether an individual prediction is stable across near-optimal models, whether instability clusters in specific regions of the input space, or what modeling choices drive it. As a result, users of automated decision systems cannot assess how reliable a prediction truly is, nor identify which subpopulations are most affected.

The problem addressed in this thesis is to characterize and quantify how predictions vary across sets of equally accurate models, with a focus on the *spatial structure* of this variation in feature space. Specifically, we ask: (i) how much do individual predictions vary across near-optimal models? (ii) does this variation cluster spatially, forming identifiable hotspots? (iii) what are the drivers—model family, hyperparameters—of this instability? and (iv) can the affected regions be described by interpretable rules?

1.4 Research Questions

Building on the problem described above, this thesis investigates the following research questions:

1. **Prediction-Level Multiplicity.** How much do individual predictions vary across equally accurate models, and which individuals are most affected? Is this variation statistically significant compared to a chance baseline?
2. **Spatial Structure of Multiplicity.** Does predictive variance exhibit spatial autocorrelation in feature space? Can we identify contiguous hotspot regions—localized clusters of high disagreement (formally defined as High–High regions via LISA in Section 3.6)—where models systematically disagree?
3. **Stability of Spatial Patterns.** Are the identified hotspot regions stable across different data splits and Rashomon set realizations, or are they artifacts of specific random seeds?
4. **Drivers of Multiplicity.** How much of the predictive variance is explained by model family versus hyperparameters within each family? Do the drivers differ between hotspot and non-hotspot regions?
5. **Robustness and Sensitivity.** Are the spatial multiplicity findings robust to (a) the Rashomon set size K , (b) the neighborhood size k in the spatial graph, and (c) post-hoc probability calibration?

6. **Validation.** Does the pipeline correctly recover known ambiguous regions when applied to synthetic data with ground-truth instability?
7. **Interpretable Characterization.** Can the identified high-multiplicity regions be described by simple, interpretable rules in terms of the original features?

1.5 Contributions of This Thesis

This thesis makes the following contributions:

1. **Spatial multiplicity framework.** We propose a pipeline that combines observation-wise predictive variance with spatial autocorrelation statistics (detailed in Chapter 3) to detect, localize, and test for significance of multiplicity hotspots in feature space.
2. **Variance decomposition.** We decompose predictive variance into between-family and within-family (hyperparameter) components, identifying which modeling choices drive instability globally and within hotspots specifically.
3. **Comprehensive empirical evaluation.** We evaluate the framework on three benchmark datasets (COMPAS, German Credit, Breast Cancer) with 10 outer runs each, five model families, and 250 candidate models per run, providing robust evidence across domains.
4. **Validation on synthetic data.** We demonstrate on synthetic datasets with known ambiguous regions that LISA HH hotspots recover the ground-truth instability regions with high precision and recall.
5. **Robustness analysis.** We show that the findings are robust to the Rashomon set size, neighborhood graph density, and post-hoc calibration, and that spatial structure is not an artifact of probability miscalibration.
6. **Interpretable rules.** We show that high-multiplicity regions can be described by compact decision rules with high purity and lift, making the spatial findings actionable for practitioners.

1.6 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews related work on predictive multiplicity, Rashomon sets, and spatial statistics. Chapter 3 presents the methodology: observation-wise variance, spatial analysis, null models, variance decomposition, and calibration robustness checks. Chapter 4 describes the experimental setup: datasets, model families, training protocol, and analysis parameters. Chapter 5 presents the results, organized by research question. Chapter 6 discusses the findings, limitations, and implications. Chapter 7 concludes and outlines directions for future work.

2 Background and Related Work

Predictive multiplicity refers to the phenomenon whereby multiple models achieve nearly identical performance yet produce meaningfully different predictions for the same individual. Recent work shows this behavior is common in realistic pipelines and has direct implications for reliability, fairness, and interpretability (Black et al., 2022). This chapter reviews the literature most relevant to this thesis, organized by theme, and identifies the methodological gaps that our spatial-variance framework addresses.

2.1 Rashomon Sets and Predictive Multiplicity

The modern formulation of multiplicity is rooted in the *Rashomon effect* named by Breiman (2001): many models can explain the same data equally well. Let $\mathcal{F}$ be a hypothesis class, $L(f)$ a loss function, and $L^* = \min_{f \in \mathcal{F}} L(f)$. The *Rashomon set* is

$$\mathcal{R}_\epsilon = \{f \in \mathcal{F} : L(f) \leq L^* + \epsilon\}.$$

This definition shifts attention from a single best model to a *set* of near-optimal models. In high-dimensional tabular settings this set can be large and structurally diverse (Semenova et al., 2022).

Rudin et al. (2024) argue that when many near-optimal models exist, model selection should explicitly weigh secondary criteria—interpretability, fairness, robustness—rather than treating one selected model as uniquely correct. Semenova et al. (2022) formalize the *Rashomon ratio* (volume of $\mathcal{R}_\epsilon$ relative to the hypothesis space) and show that large ratios imply the existence of simpler, interpretable models at little accuracy cost.

Marx et al. (2020) coined the term *predictive multiplicity* and introduced the first formal measures—*ambiguity* (maximum fraction of individuals whose predicted label can be flipped) and *discrepancy* (maximum pointwise prediction difference)—for linear classifiers. Black et al. (2022) subsequently broadened the scope, framing model multiplicity as a systemic challenge for accountability: if equally valid models disagree on individual decisions, which model’s output should govern?

2.2 Disagreement Metrics: From Labels to Probabilities

Label-space metrics. The ambiguity and discrepancy of Marx et al. (2020) operate on binary labels relative to a baseline model. While intuitive, these metrics collapse probability information to a 0/1 threshold, potentially masking calibration-relevant variation.

Probability-space metrics. Watson-Daniels et al. (2023) extend predictive multiplicity to *probabilistic classification*, defining measures that capture variation in risk scores (predicted probabilities) rather than hard labels. Their framework quantifies both the prevalence of conflicting risk scores and their severity across near-optimal convex models, and relates multiplicity to dataset characteristics such as separability and outlier structure. In parallel, Hsu and Calmon (2022) propose *Rashomon Capacity*, a global, information-theoretic measure of how broadly the Rashomon set spans prediction space. While Rashomon Capacity captures diversity at the dataset level, it does not resolve *where* in feature space disagreement concentrates.

Per-instance metrics and the conflict ratio. Per-instance feasible ranges provide finer resolution. Hsu et al. (2024) introduce the *Viable Prediction Range* (VPR) for gradient boosting, estimating for each observation the min/max probability attainable inside $\mathcal{R}_\epsilon$. The *conflict ratio* $c(x) = \min(q(x), 1 - q(x))$, where $q(x)$ is the fraction of models predicting above a decision threshold, measures per-point label disagreement on a $[0, 0.5]$ scale. Frohnapfel et al. (2026) argue that individual conflict ratios should be reported to comply with the EU AI Act’s provisions on per-person accuracy, making the metric directly policy-relevant.

Procedural instability. Orthogonal to the Rashomon-set perspective, Cooper et al. (2024) study *self-consistency* across repeated training runs of the same pipeline, quantifying disagreement induced by random seeds and stochasticity even within one model family. Watson-Daniels et al. (2025) connect this to *predictive churn*—the instability of individual predictions when a deployed model is retrained—showing theoretical and empirical links between churn and Rashomon-set multiplicity.

Gap. All of the above metrics are either global summaries or pointwise scalars; none identify *spatially contiguous regions* of instability in feature space, nor do they decompose disagreement by its source (model family vs. hyperparameter).

2.3 Variable Importance and Source Attribution Across Models

Classical variable importance is model-specific: a feature is “important” for one fitted model but may be unimportant for another equally accurate model. Fisher et al. (2019) address this with *Model Class Reliance* (MCR), computing the range of permutation-importance values across all models in $\mathcal{R}_\epsilon$. Dong and Rudin (2020) extend this idea to *Variable Importance Clouds*, visualizing the joint distribution of importance scores for every feature across the set of good models. Donnelly et al. (2024) propose the *Rashomon Importance Distribution* (RID), a framework that quantifies feature importance across $\mathcal{R}_\epsilon$ with consistency guarantees under data perturbation.

These lines of work attribute variation in *explanations* across models to individual features. Our thesis pursues a complementary direction: attributing variation in *predictions* (i.e., multiplicity) to model families and, within each family, to hyperparameters—a variance-decomposition approach that quantifies structural sources of disagreement rather than feature-level explanations.

Hyperparameter role. Cavus et al. (2025) provide the closest antecedent to our hyperparameter analysis. Studying six model families on 21 benchmark datasets, they show that tuning key hyperparameters (e.g., regularization strength, tree depth, learning rate) significantly increases prediction discrepancy within each family. Our work builds on this insight by decomposing per-observation variance into between-family and within-family (hyperparameter) components, and by identifying which hyperparameters contribute most to disagreement inside spatial hotspots.

2.4 Local Structure and Hotspot Detection

Spatial statistics are well established for detecting non-random local structure in geospatial and networked data. *Global Moran’s I* tests whether a continuous variable exhibits overall spatial autocorrelation under a neighborhood graph, while *Local Indicators of Spatial Association* (LISA) decompose this into per-observation statistics that identify *High–High* (HH) and *Low–Low* (LL) clusters. Our key methodological transfer is to replace the geographic adjacency matrix with a k -nearest-neighbor (kNN) graph in feature space, treating per-observation prediction variance as the spatial variable.

In the multiplicity literature, explicit feature-space localization remains underdeveloped. Prior work overwhelmingly quantifies *how much* models disagree, not *where* they disagree in a contiguous, statistically testable way. Watson-Daniels et al. (2023) observe that multiplicity concentrates near decision boundaries and among outliers, but do not formalize localization via spatial statistics. Herbinger et al. (2022) detect regions with heterogeneous feature effects (REPID), which is conceptually related—both seek local structure—but targets single-model explanations rather than cross-model disagreement. This thesis operationalizes the missing localization step: we apply Moran’s I and LISA to per-point variance and conflict across the Rashomon set, yielding statistically significant HH hotspots that can be mapped, compared across seeds, and characterized by interpretable rules.

2.5 Interpretable Characterization of Instability Regions

Interpretability research typically explains the decision boundary of a *single* model. Under multiplicity the central object is not one boundary but a *region of instability* across many near-optimal models. Kobylińska et al. (2023) propose RASHOMON_DETECT to identify models in the Rashomon set that behave differently, using a *Profile Disparity Index* based on partial-dependence profiles. Their goal is to flag heterogeneous explanations; our goal is to flag heterogeneous *predictions at the observation level* and characterize the affected subpopulations.

Regional effect methods such as REPID (Herbinger et al., 2022) and forward marginal effects (Scholbeck et al., 2024) partition feature space into regions with homogeneous feature effects. We draw on a related principle—subgroup discovery via decision rules—but apply it to a different target: the HH mask produced by LISA, rather than a single model’s residuals. The resulting rules describe *who* is most subject to model arbitrariness, not why one model predicts a particular label.

2.6 Fairness Under Model Multiplicity

Fairness analyses of multiplicity are a growing but still nascent area. Sokol et al. (2024) define *cross-model fairness*: an individual harmed when one predictor is chosen ad hoc from a pool of equally performing models could argue for the model that grants the most favorable outcome. Their empirical study on real-world datasets confirms that such unfairness is pervasive and difficult to eliminate by technical means alone.

Long et al. (2023) demonstrate that optimizing for group-fairness metrics (e.g., statistical parity) while ignoring multiplicity can *increase* individual arbitrariness, masking high

predictive multiplicity behind favorable aggregate fairness scores. Dai et al. (2025) complement this by studying the Rashomon set’s properties from a fairness perspective: they derive efficient sampling methods, estimate individual flip probabilities, and show that intentionally searching for fair models within $\mathcal{R}_\epsilon$ is far more effective than hoping arbitrary selection will suffice. Cooper et al. (2024) make a similar point empirically, finding that most fair-classification benchmarks appear close-to-fair once prediction variance is taken into account.

Our thesis does not propose new fairness constraints; instead, it provides a descriptive diagnostic—subgroup exposure analysis—that quantifies whether HH hotspots disproportionately concentrate in specific demographic groups, adding a practically relevant layer to multiplicity audits.

2.7 Robustness, Calibration, and Mitigation

Robustness of scientific and predictive conclusions to analyst choices is formalized by Coker et al. (2021) through *hacking intervals*—the range of a summary statistic obtainable under plausible data-analysis manipulations. In the multiplicity context, the analogous concern is whether spatial patterns (Moran’s I , HH hotspot locations) are robust to incidental choices such as Rashomon-set size K , kNN neighborhood k , and probability calibration.

Calibration is particularly relevant because Platt scaling or isotonic regression can alter predicted probabilities without changing model ranking. If hotspots disappear after calibration, they may reflect probability miscalibration rather than genuine structural disagreement. Our calibration-robustness analysis (Section 5.9) tests this directly, finding that HH hotspot locations are largely preserved after Platt scaling, with high Jaccard overlap between pre- and post-calibration masks.

Haghighat et al. (2026) propose three mitigation strategies—outlier correction, local patching, and pairwise reconciliation—to reduce disagreement within the Rashomon set while maintaining accuracy. Their approach is complementary to ours: whereas they aim to *resolve* multiplicity (make models agree), we aim to *diagnose* it (identify where and why models disagree) as a prerequisite for informed intervention.

2.8 Positioning of This Thesis

Relative to prior work, this thesis contributes by integrating five strands that are usually studied in isolation:

1. **Rashomon-set predictive multiplicity** (Marx et al., 2020, Black et al., 2022), operationalized with both variance and conflict metrics at the per-observation level;
2. **Spatial localization** via Moran’s I and LISA applied to feature-space kNN graphs, transferring tools from spatial statistics to detect statistically significant hotspots of disagreement;
3. **Source attribution** via hierarchical variance decomposition—first between model families, then within each family by hyperparameter—building on the insights of Cavus et al. (2025);

4. **Interpretable characterization** of instability regions through decision-rule extraction for HH hotspots, connecting multiplicity diagnosis to actionable subpopulation descriptions;
5. **Robustness diagnostics** covering sensitivity to K , k , calibration, and cross-seed stability of hotspot locations.

The resulting perspective treats predictive multiplicity as a *localized, structured phenomenon* rather than a single global nuisance metric, and provides diagnostics that are directly usable for model governance in high-stakes settings.

3 Methodology

This chapter presents the methodology used to analyze predictive multiplicity through a spatial lens. The proposed approach characterizes how predictions vary across a set of near-optimal models and investigates whether this variation exhibits statistically significant structure in feature space. The methodology is independent of any particular dataset or experimental configuration; dataset-specific choices are described in Chapter 4.

3.1 Problem Setting and Notation

Let (X, y) denote a supervised learning dataset with N observations, where $X \in R^{N \times d}$ is the feature matrix and $y \in \{0, 1\}^N$ is a binary target variable. Let $\mathcal{F}$ denote a hypothesis class of predictive models, potentially spanning multiple model families and hyperparameter configurations.

We consider probabilistic classifiers $f \in \mathcal{F}$ that output predicted probabilities $\hat{p}_{fi} \in [0, 1]$ for each observation $i \in \{1, \dots, N\}$. Model performance is measured using a loss function $L(f)$ evaluated on held-out data.

3.2 Rashomon Set of Near-Optimal Models

Predictive multiplicity is studied through the Rashomon set, defined as the collection of models whose performance lies within a tolerance $\varepsilon > 0$ of the best achievable loss. In our setting, $\mathcal{F}$ is a *finite* candidate pool: a set of models trained with varying families and hyperparameter configurations on a common training set. Let

$$L^* = \min_{f \in \mathcal{F}} L(f)$$

denote the minimum validation loss (Brier score) among all candidates. The Rashomon set is then

$$\mathcal{R}_\varepsilon = \{f \in \mathcal{F} : L(f) \leq L^* + \varepsilon\}.$$

The tolerance ε controls the strictness of selection. Because the exact value of ε that produces a given set size varies across datasets and splits, we adopt an equivalent **top- K approximation**: the K models with the lowest validation Brier score are selected, yielding $|\mathcal{R}| = K$. This fixes the Rashomon set size across runs and datasets, making results comparable. Selection can be *global* (top- K across all families) or *per-family* (top- K within each model family); the latter preserves enough within-family variation for hyperparameter decomposition (Section 3.10). Sensitivity to K is examined empirically.

3.3 Observation-Wise Predictive Variance

To quantify predictive multiplicity at the level of individual observations, we compute the variance of predicted probabilities across the Rashomon set. Let $|\mathcal{R}_\varepsilon| = M$ denote the number of models in the Rashomon set, and let $\hat{p}_{mi}$ denote the predicted probability for observation i produced by model $m \in \mathcal{R}_\varepsilon$.

The observation-wise predictive variance is defined as

$$v_i = \text{Var}_{m \in \mathcal{R}_\varepsilon}(\hat{p}_{mi}), \quad i = 1, \dots, N.$$

This yields a scalar instability score for each observation, capturing the magnitude of disagreement among near-optimal models. Unlike range-based or binary disagreement metrics, predictive variance reflects the full distribution of predictions within the Rashomon set.

3.4 Observation-Wise Conflict Ratio

As a complementary measure of disagreement, we define the *conflict ratio* for each observation. Let $\tau = 0.5$ be a decision threshold and let

$$q_i = \frac{1}{M} \sum_{m=1}^M \mathbf{1}[\hat{p}_{mi} \geq \tau]$$

denote the fraction of Rashomon models that predict a positive label for observation i . The conflict ratio is

$$c_i = \min(q_i, 1 - q_i), \quad c_i \in [0, 0.5].$$

A value of $c_i = 0$ indicates unanimous agreement on the predicted label; $c_i = 0.5$ indicates a maximally split ensemble. Because c_i captures hard-label disagreement it is directly related to the variance of binary votes: $\text{Var}(\hat{y}_{mi}) = q_i(1 - q_i) = c_i(1 - c_i)$ when computed across models (Frohnepfel et al., 2026).

We apply the same spatial analysis (Moran’s I , LISA; Sections 3.6–3.7) to c_i as to v_i , yielding conflict-based hotspot regions that identify areas of concentrated label disagreement.

3.5 Feature-Space Graph Construction

To study the spatial organization of predictive variance, we represent the dataset as a graph in feature space. Observations are treated as nodes, and edges encode local neighborhood relationships.

A k -nearest-neighbor (kNN) graph is constructed using the (preprocessed) feature vectors in X with Euclidean distance. The initial graph is *directed*: observation i has an outgoing edge to each of its k nearest neighbors. We then symmetrize the adjacency so that an edge exists between i and j whenever *either* i is among the k nearest neighbors of j *or* j is among the k nearest neighbors of i . Let $A \in \{0, 1\}^{N \times N}$ denote the resulting symmetric binary adjacency matrix ($A_{ij} = 1$ if i and j are connected, $A_{ii} = 0$).

The spatial weight matrix W is obtained by *row-standardizing* A :

$$W_{ij} = \frac{A_{ij}}{\sum_{j'=1}^N A_{ij'}}, \quad i = 1, \dots, N.$$

Each row of W sums to one, so the *spatial lag* of a variable $\mathbf{v}$ at observation i is the weighted average of its neighbors’ values:

$$[W\mathbf{v}]_i = \sum_{j=1}^N W_{ij} v_j.$$

This spatial lag enters both global Moran’s I and the local LISA statistics defined below.

3.6 Global Spatial Autocorrelation

To test whether predictive variance is spatially structured in feature space, we compute the global Moran’s I statistic. Let $\mathbf{v} = (v_1, \dots, v_N)^\top$ denote the vector of observation-wise predictive variances, and let $\bar{v}$ denote its mean.

Global Moran’s I is defined as

$$I = \frac{\sum_{i=1}^N \sum_{j=1}^N W_{ij} (v_i - \bar{v})(v_j - \bar{v})}{\sum_{i=1}^N (v_i - \bar{v})^2}.$$

Positive values of I indicate that observations with similar levels of predictive variance tend to be neighbors in feature space, while values near zero suggest spatial randomness. Statistical significance of global Moran’s I is assessed via a permutation test: the variance values v_i are randomly reassigned across observations while the spatial weight matrix W is held fixed, yielding a null distribution for I . (This global-level permutation test is distinct from the observation-level permutation null described in Section 3.9, which permutes model predictions rather than variance scores.)

3.7 Local Indicators of Spatial Association

While global Moran’s I assesses overall spatial dependence, it does not identify *where* such dependence occurs. To localize regions of elevated predictive multiplicity, we compute Local Indicators of Spatial Association (LISA).

For each observation i , let $z_i = v_i - \bar{v}$ denote the mean-centered predictive variance and let $l_i = [W\mathbf{z}]_i = \sum_j W_{ij} z_j$ denote its spatial lag. The local Moran statistic is

$$I_i = z_i \cdot l_i.$$

Significance testing. For each observation a *conditional* permutation test is performed: the values $\{z_j : j \neq i\}$ are randomly permuted (999 times by default) while z_i and W are held fixed, yielding a reference distribution for I_i . A pseudo p -value is computed as $(1 + \#\{\text{null} \geq \text{observed}\}) / (R_{\text{local}} + 1)$. To control the family-wise false discovery rate across all N tests, the Benjamini–Hochberg (BH) procedure is applied at level $\alpha = 0.05$.

Quadrant classification. Each observation whose adjusted p -value falls below α is classified into one of four categories based on the signs of z_i and l_i :

- **High–High (HH):** $z_i > 0$ and $l_i > 0$ — the observation has above-average variance *and* its neighbors have above-average variance. These are the *multiplicity hotspots*.
- **Low–Low (LL):** $z_i < 0$ and $l_i < 0$ — both the observation and its neighborhood have below-average variance (stable coldspots).
- **High–Low (HL):** $z_i > 0$, $l_i < 0$ — a high-variance outlier surrounded by low-variance neighbors.

- **Low–High (LH):** $z_i < 0$, $l_i > 0$ — a low-variance observation in a high-variance neighborhood.

Observations that do not reach significance after BH correction are labeled *non-significant*. Throughout this thesis, “HH region” refers specifically to the set of observations classified as High–High by this procedure.

3.8 Connected Components of High-Multiplicity Regions

To extract coherent regions of predictive multiplicity, we restrict the kNN graph to observations classified as High–High and identify connected components within this subgraph. Each connected component represents a contiguous region of feature space in which predictive variance is consistently elevated. Components below a minimum size threshold may be discarded to avoid spurious detections. The resulting components support region-level analysis and stability assessments across different data splits or Rashomon set realizations.

3.9 Null Models for Spatial Multiplicity

To ensure that detected spatial structure is not an artifact of marginal variance distributions, we employ null models based on permutation of predictions.

The null transformation operates on the prediction matrix $P \in R^{M \times N}$. For each model m independently, the vector of predicted probabilities $(\hat{p}_{m1}, \dots, \hat{p}_{mN})$ is randomly permuted across observations. This *within-model permutation* has two key properties:

1. **Preserved:** the marginal distribution of each model’s predictions (i.e., the set of predicted values is unchanged, only their assignment to observations differs).
2. **Destroyed:** the alignment between any observation’s predictions across models and its position in feature space. In particular, any spatial clustering of the resulting observation-wise variance v_i or conflict ratio c_i is due to chance alone.

For each of R permutations, observation-wise variance (and, separately, conflict ratio) and global Moran’s I are recomputed on the permuted data. The empirical p -value for the observed statistic is

$$p = \frac{1 + \#\{\text{null} \geq \text{observed}\}}{R + 1}.$$

This design tests specifically whether spatially structured disagreement exceeds what would arise from independent, spatially unstructured predictions of the same overall magnitude.

3.10 Hyperparameter Variance Decomposition

To investigate the *sources* of predictive multiplicity, we decompose disagreement among Rashomon models into contributions attributable to model family and, within each family, to individual hyperparameters. Because hyperparameters are family-specific (e.g., tree depth applies only to tree-based models), mixing families would introduce missing values and spurious importance. The analysis is therefore conducted *per family*.

Per-family Rashomon selection. For each model family $\mathcal{F}_j$ (e.g., GBM, RF, MLP), we select the top- K models by validation Brier score within that family, yielding a within-family Rashomon set of size $K_j = \min(K, |\mathcal{F}_j|)$.

Model-level disagreement contribution. Let $P^{(j)} \in R^{K_j \times N_{\text{test}}}$ denote the test predictions of the within-family Rashomon set, and let

$$\bar{p}_i^{(j)} = \frac{1}{K_j} \sum_{m=1}^{K_j} \hat{p}_{mi}^{(j)}$$

be the family-ensemble mean for observation i . The *disagreement contribution* of model m is

$$V_m = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (\hat{p}_{mi}^{(j)} - \bar{p}_i^{(j)})^2. \quad (1)$$

V_m measures the mean squared deviation of model m 's predictions from its family ensemble; higher values indicate that the model contributes more to within-family disagreement.

HP importance via between-group variance of V_m . For a hyperparameter h with G distinct values (after dropping missing or constant columns), we partition the K_j models into groups $\{S_1, \dots, S_G\}$ by their value of h . Standard one-way variance decomposition yields

$$\underbrace{\text{Var}(V_m)}_{\text{total}} = \underbrace{\sum_{g=1}^G \frac{|S_g|}{K_j} (\bar{V}_{S_g} - \bar{V})^2}_{\text{Var}_{\text{between}}(h)} + \underbrace{\sum_{g=1}^G \frac{|S_g|}{K_j} \text{Var}_{m \in S_g}(V_m)}_{\text{Var}_{\text{within}}(h)},$$

where $\bar{V}$ is the overall mean and $\bar{V}_{S_g}$ the group mean. The *importance* of hyperparameter h is

$$\rho(h) = \frac{\text{Var}_{\text{between}}(h)}{\text{Var}_{\text{total}}}. \quad (2)$$

This ratio-of-variances formulation is preferred over a mean-of-per-observation-ratios because it is more stable when individual observations have near-zero total variance.

Edge cases. Hyperparameter columns with all missing values or fewer than two unique values (after dropping NaN) are excluded. Groups with fewer than one model are dropped before computing the decomposition. If after filtering fewer than two groups remain, the hyperparameter is skipped.

3.11 Calibration Robustness Check

Observation-wise variance v_i is computed from raw predicted probabilities $\hat{p}_{mi}$, which may be poorly calibrated. If spatial clustering were an artifact of systematic miscalibration (e.g., consistently over-confident predictions near certain feature regions), correcting calibration should substantially reduce Moran's I and shift HH regions. We therefore apply a post-hoc calibration step and compare the resulting multiplicity and spatial metrics to their uncalibrated counterparts.

Calibration procedure. For each model m in the Rashomon set, a *Platt scaling* model is fitted: a logistic regression $\sigma(a\hat{p} + b)$ is trained on the validation set $(\hat{p}_{m,\text{val}}, y_{\text{val}})$, where $\hat{p}_{m,\text{val}}$ are the model’s validation-set predicted probabilities and y_{val} are the true labels. The fitted mapping is then applied to obtain calibrated test predictions $\tilde{p}_{mi} = \sigma(a\hat{p}_{mi} + b)$ for each test observation i . Crucially, the calibration mapping is fit on the validation set only, so no test-set information leaks into the calibrated predictions.

Before/after comparison. The full multiplicity and spatial pipeline — mean variance, conflict ratio, Moran’s I , and LISA classification — is recomputed on the calibrated prediction matrix $\tilde{P}$. We record the change in each metric (Δ mean variance, Δ Moran’s I , Δ HH count) and the Jaccard similarity between the HH masks before and after calibration:

$$J_{\text{cal}} = \frac{|\text{HH}_{\text{before}} \cap \text{HH}_{\text{after}}|}{|\text{HH}_{\text{before}} \cup \text{HH}_{\text{after}}|}.$$

High J_{cal} and small Δ Moran’s I indicate that spatial structure is robust to calibration and reflects genuine prediction disagreement rather than probability-scale artifacts.

3.12 Validation on Synthetic Data

To validate that LISA HH regions correspond to genuinely ambiguous regions in feature space, we evaluate on synthetic datasets with *known* ground-truth ambiguous regions. These datasets combine well-separated stable regions (where models tend to agree) with designed “islands” where the conditional probability $P(y = 1|x)$ is near 0.5, so that near-optimal models can disagree. Recovery of these ground-truth regions by the HH classification (e.g. precision, recall, Jaccard overlap) provides a quantitative validation criterion for the spatial multiplicity pipeline.

3.13 Interpretable Rule Extraction

Once high-multiplicity regions have been identified via LISA (HH hotspots) or thresholding (top quantile of pointwise variance), a natural question is whether these regions can be described by simple, interpretable rules in terms of the original features. We frame this as a supervised classification task: label each test observation as belonging to the high-multiplicity region or not, then fit interpretable models to recover the labeling.

Four complementary methods are employed:

1. **Shallow decision trees.** A depth-limited decision tree is fitted to the binary label (HH or HV vs. rest). Each leaf defines a conjunctive rule; rules are evaluated by *purity* (fraction of positives in the rule region) and *support* (number of observations matched).
2. **L1-regularized logistic regression.** A logistic regression with L1 penalty identifies features with nonzero coefficients that are associated with high-multiplicity membership. The coefficient magnitudes indicate feature importance.

3. **Beam search for conjunctive rules.** A greedy beam search constructs conjunctions of feature conditions that maximize *lift* (purity divided by base rate), subject to a minimum support constraint.
4. **Cluster-describe.** DBSCAN clustering of the high-multiplicity points in feature space identifies spatial clusters, and a decision tree is fitted to each cluster to produce cluster-specific rules.

Rules are evaluated using purity, support, lift, and out-of-bag (OOB) stability under bootstrap resampling. Permutation enrichment tests assess whether the observed purity exceeds what would be expected by chance.

3.14 Decision Boundary Proximity Analysis

To test whether high-multiplicity regions are simply those near the decision boundary, we compute the margin $m_i = |\bar{p}_i - 0.5|$ for each test observation, where $\bar{p}_i$ is the ensemble mean predicted probability. We then compute the Pearson and Spearman correlations between pointwise variance and margin, and compare mean margins for HH vs. non-HH points using the Mann–Whitney U test.

3.15 Fairness and Subgroup Exposure

To assess whether HH and HV regions disproportionately affect demographic subgroups, we compute group-wise exposure rates: the fraction of each group’s test points classified as HH (or HV), with uncertainty summarized by bootstrap confidence intervals across runs. For inferential reporting, we apply two safeguards: (i) per-run subgroup minimum size filtering, and (ii) minimum cross-run support for each subgroup. Significance is then evaluated with a *stratified* permutation test: labels are shuffled within each run (preserving each run’s sample composition), run-level rates are recomputed, and the aggregated inter-group disparity (max–min HH rate across eligible groups) is compared to its null distribution.

3.16 Alternative Graph Constructions

To verify that spatial results are not artifacts of the Euclidean distance metric on the full one-hot-encoded feature space, we compare the baseline kNN graph with two alternatives: (1) kNN in PCA-reduced space (retaining 15 components), and (2) kNN using cosine distance on L2-normalized features. All other pipeline parameters remain fixed; Moran’s I and HH counts are compared across methods.

3.17 Summary

The methodology described in this chapter provides a unified framework for analyzing predictive multiplicity as a spatial phenomenon. By combining Rashomon set analysis, observation-wise predictive variance and conflict ratio, spatial statistics (Moran’s I , LISA), null models, hyperparameter variance decomposition, calibration checks, validation on synthetic data, interpretable rule extraction, decision boundary proximity analysis,

fairness exposure assessment, and alternative graph constructions, the approach enables identification, localization, interpretation, and characterization of regions in feature space where model predictions are particularly unstable, along with robustness verification of the findings.

4 Experimental setup

This chapter describes how the methodology introduced in Chapter 3 is instantiated in the experiments. We specify the datasets, model families, training protocol, Rashomon set construction, and spatial analysis parameters. All design choices are fixed across experiments unless stated otherwise.

4.1 Datasets

Experiments are conducted on three benchmark datasets commonly used in fairness, interpretability, and predictive uncertainty research:

- **COMPAS:** The ProPublica recidivism dataset. Features include age, prior offence count (numeric) and sex, race, charge degree (categorical). The binary target indicates whether the individual reoffended within two years.
- **German Credit:** The credit-g dataset from OpenML (version 1). It contains financial and demographic attributes; the target is good vs. bad credit risk (binary). Numeric and categorical features are determined automatically from dtypes.
- **Breast Cancer Wisconsin:** The breast cancer dataset from scikit-learn, with numeric features derived from cell nuclei measurements and a binary malignancy label.

All tasks are binary classification with a mix of numeric and categorical features. Preprocessing is described below.

4.2 Data Splitting

Each dataset is partitioned into training (60%), validation (20%), and test (20%) sets using stratified sampling. Two splitting modes are used depending on the analysis goal:

Primary (independent) splits. Each outer seed produces a fully independent stratified split; all three sets differ between seeds. This mode is used for the main multiplicity and spatial analyses reported in Sections 5.1–5.8, where we require unbiased variability estimates across runs.

Fixed-test-set splits. The test set is drawn once (seed 0) and held fixed across all outer seeds; only the train/validation boundary is re-split per seed. This mode is used for cross-seed HH stability and Jaccard overlap analyses (Section 5.5), where point-level comparison across seeds is meaningful only if the same individuals appear in every test set.

In both modes, the test set is held out from all model training and Rashomon set construction. It is used only for generating predictions from the selected models, computing observation-wise metrics, and performing spatial and null analyses.

4.3 Model Families

Models are trained across five model families to induce diverse predictive multiplicity:

- **Logistic Regression (LogReg)**: Elastic net penalty (`penalty='elasticnet'`, `solver='saga'`), with `C`, `l1_ratio`, and `max_iter` varied.
- **k -Nearest Neighbors (kNN)**: `n_neighbors`, `weights` (uniform/distance), and `p` (1 or 2 for Minkowski distance) varied.
- **Random Forest (RF)**: `n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features`; `bootstrap=True`.
- **Gradient Boosting (GBM)**: `n_estimators`, `learning_rate`, `max_depth`, `subsample`.
- **Multilayer Perceptron (MLP)**: `hidden_layer_sizes`, `alpha`, `learning_rate_init`, `activation` (ReLU/tanh); `solver='adam'`, `early_stopping=True`.

These families span linear, instance-based, tree-based, and neural network models.

4.4 Hyperparameter Sampling and Training

For each model family, 50 hyperparameter configurations are sampled uniformly at random from the predefined search spaces. For families with few discrete options (e.g. LogReg, kNN), configurations are deduplicated and resampled until 50 unique configurations are obtained. Each configuration is trained with a deterministic seed derived from the run identifier to ensure reproducibility. Training uses the training set only; the validation set is used exclusively for evaluating candidate models.

Model performance is evaluated using the *Brier score* (mean squared error between predicted probabilities and binary labels) on the validation set. No cross-validation is used in the main pipeline: each outer run uses a single train/validation/test split.

4.5 Preprocessing

Preprocessing is applied uniformly across all model families via a column-transformer fitted on the training set only.

- **Numeric features**: missing values are imputed with the training-set median, then standardized to zero mean and unit variance.
- **Categorical features**: missing values are imputed with the training-set mode, then one-hot encoded (unknown categories at test time are mapped to a zero vector).
- **Remaining columns**: any column not identified as numeric or categorical is dropped.

The fitted transformer is applied identically to validation and test data without refitting, ensuring no data leakage. The same transformed feature space is used both for model training and for constructing the k NN graph (Section 4.8), so that spatial neighborhoods reflect the distances models actually operate on.

4.6 Rashomon Set Construction

The Rashomon set is operationalized by selecting the **top- K** models by validation Brier score, with $K = 25$ unless otherwise stated. This yields a fixed-size set of near-optimal models per run and ensures comparability across runs and datasets. Two selection modes are supported:

- **Global:** The 25 models with lowest validation Brier score are selected from the full candidate pool (all families combined).
- **Per-family (total- K):** For fair global-vs-family comparisons, the total budget $K = 25$ is distributed across families (approximately equally, with deterministic tie-breaking by best family-level validation Brier).
- **Per-family (K -each):** For within-family driver analyses, the top 25 models are selected within each family (up to 125 models total) to preserve enough within-family variation for hyperparameter decomposition.

Unless otherwise stated, the global selection mode is used. Sensitivity of multiplicity and spatial metrics to K is examined by varying $K \in \{5, 10, 15, 20, 25, 30, \dots\}$ in separate analyses. Results are reported for **both** the global Rashomon set and per-family selections: total- K per-family selection is used for fair head-to-head comparisons with global- K , while K -each per-family selection is used when the objective is within-family decomposition (see Sections 5.4 and 5.8).

4.7 Prediction Matrix and Multiplicity Metrics

For each run, all models in the Rashomon set are applied to the held-out test set to obtain predicted probabilities. This yields a prediction matrix $P \in R^{M \times N}$, where M is the number of selected models and N is the number of test observations. Observation-wise predictive variance is computed from P as in Section 3.3. Additional multiplicity metrics include mean variance, ambiguity (mean over observations of $\max_m \hat{p}_m - \min_m \hat{p}_m$), disagreement rate (fraction of test points where two models differ by more than a threshold $\delta = 0.05$), and discrepancy (maximum over model pairs of mean absolute prediction difference). These ambiguity/discrepancy definitions follow the probability-space formulation used in the *Resolving Predictive Multiplicity for the Rashomon Set* framework.

4.8 Feature-Space Graph and Spatial Analysis

The spatial weight matrix W is built from a k -nearest-neighbor (kNN) graph in feature space. Features are the preprocessed test-set vectors (same as used for model predictions). The default neighborhood size is $k = 30$; sensitivity to k is examined with $k \in \{10, 20, 30, 50\}$. Distances are Euclidean; the adjacency matrix is row-standardized. Global Moran’s I is computed with 999 permutations for significance. Local Moran (LISA) is computed with 999 permutations per observation; the Benjamini–Hochberg procedure is applied at $\alpha = 0.05$ to control the false discovery rate when labeling High–High (HH) and Low–Low (LL) clusters. Observations classified as HH are interpreted as locally unstable

regions. Connected components of HH observations are extracted from the restricted kNN graph; a minimum component size may be applied to filter small clusters.

4.9 Null Model Configuration

Null distributions are constructed by permuting each model’s predictions independently across observations (within-model permutation). Concretely, for every model m the vector of predicted probabilities $(\hat{p}_{m1}, \dots, \hat{p}_{mN})$ is randomly shuffled so that each $\hat{p}_{mi}$ is reassigned to a different observation. This preserves the marginal distribution of predictions for each model while destroying any alignment between predictive variance and feature-space location.

For each of $R = 100$ permutations, observation-wise variance (and, separately, conflict ratio) and global Moran’s I are recomputed on the permuted data. The empirical p -value for the observed statistic is

$$p = \frac{1 + \#\{\text{null} \geq \text{observed}\}}{R + 1}.$$

A run is considered to show significant spatial structure if $p < 0.05$. With $R = 100$ the smallest attainable p -value is $1/101 \approx 0.0099$; consequently, observed p -values reported as “ $p < 0.01$ ” should be interpreted as being at the resolution floor of the permutation test rather than as an exact probability.

4.10 Hyperparameter Variance Decomposition

The HP importance analysis described in Section 3.10 is instantiated as follows. For each outer seed and each model family, the top- $K = 25$ models by validation Brier score within that family are selected (per-family K -each mode; see Section 4.6). If a family has fewer than K candidates in a given run, all available candidates are used and the effective K_{actual} is logged.

For each selected model, the disagreement contribution V_m (Eq. 1) is computed from test-set predictions. Hyperparameters are parsed from the per-model configuration stored alongside each run; columns with fewer than two unique values within the family are dropped. HP importance $\rho(h)$ (Eq. 2) is computed for each remaining hyperparameter via one-way variance decomposition of V_m .

This procedure is repeated independently for all 10 outer seeds. Results are aggregated across seeds: we report mean importance $\pm$ standard deviation, and rank stability (frequency with which each hyperparameter appears in the top 3 across seeds). The analysis can optionally be restricted to observations classified as HH to identify hotspot-specific drivers.

4.11 Outer Runs and Reproducibility

For each dataset, 10 outer runs are performed with different random seeds (e.g. seeds 0–9). Each outer run corresponds to an independent train/validation/test split and a full training and selection pipeline. Results are aggregated across outer runs (e.g. mean and standard deviation of mean variance, Moran’s I , and fraction of runs with significant

Moran’s I). All random seeds are fixed so that experiments are reproducible. Code and configuration are provided in the electronic appendix.

4.12 Implementation

Experiments are implemented in Python using `scikit-learn` for models and preprocessing, `libpysal` and `esda` for spatial weights and Moran statistics, and `NumPy/SciPy` for numerical computation. The training pipeline saves per run: split indices, model metadata (family, hyperparameters, validation Brier score), and prediction matrices for validation and test sets. The analysis layer loads these saved artifacts and performs Rashomon selection, multiplicity metrics, spatial analysis, and null experiments without retraining. All code and configurations are available in the accompanying repository.

4.13 Calibration Robustness

For each run on the three benchmark datasets (COMPAS, German Credit, Breast Cancer), we apply Platt scaling to test whether spatial multiplicity is driven by probability miscalibration. For each model in the Rashomon set (top- K by validation Brier), a logistic regression is fitted on validation-set predicted probabilities versus validation labels; this mapping is then applied to that model’s test-set predictions. The same multiplicity and spatial pipeline (mean variance, Moran’s I , LISA with FDR, HH/LL masks, connected components) is run on both uncalibrated and calibrated test predictions. We record before/after metrics (Brier score, mean variance, Moran’s I , number of HH points, number of components) and the Jaccard similarity between HH masks before and after calibration. Results are aggregated over runs (mean $\pm$ std). Synthetic datasets are excluded from this analysis because they use a different data-generating process.

4.14 Synthetic Datasets

Two synthetic data designs are used to validate that LISA HH regions recover known ambiguous regions:

- **Single-island:** A mixture of two well-separated Gaussian blobs (stable region) and one “ambiguous island”—points drawn uniformly in a disk around the origin with a weak XOR-like conditional probability $P(y = 1|x) = 0.5 \pm \delta$. Labels are sampled from this probability. Ground truth distinguishes island observations from stable-region observations.
- **Three-islands + outliers:** Two stable blobs, three ambiguous islands (each with weak XOR signal around a center), and a small fraction of isolated outliers. Ground truth labels each point as belonging to one of the three islands, the stable region, or the outlier set.

The same training pipeline (60/20/20 split, same model families, top- K by validation Brier) is applied. Multiplicity and spatial analysis are run on the test set. Recovery of ground-truth islands by the LISA HH classification is evaluated via precision, recall, and

Jaccard overlap; sensitivity to FDR α can be examined. The null experiment (permutation of predictions, Section 4.9) is also run to confirm that observed Moran’s I is significant.

4.15 Hyperparameter and Family Importance

To attribute predictive multiplicity to model family and hyperparameters, the following analyses are conducted for each benchmark dataset across all 10 outer seeds:

- **Per-family Rashomon set:** The top- $K = 25$ models *within each* model family by validation Brier are selected (up to 125 models total, 25 per family). Observation-wise variance is decomposed into between-family and within-family components; the ratio of between-family variance to total variance (“family importance”) is computed per observation and summarized (mean, median, 90th percentile).
- **Within-family HP importance:** For each family, the model-level disagreement contribution V_m (Section 3.10) is decomposed by hyperparameter using the between-group/total ratio $\rho(h)$ (Eq. 2). Results are reported per family per seed and aggregated across seeds.
- **HH-specific importance:** The same within-family HP importance is computed *restricted to HH observations* (Section 3.7) to identify hotspot-specific drivers.
- **Family importance by subset:** Family importance is computed on three subsets—all test points, HH points only, and non-HH points only—to compare how much model family explains multiplicity in hotspot versus non-hotspot regions.

4.16 Interpretable Rule Extraction

To characterize high-multiplicity regions in terms of original features, interpretable rules are extracted on each benchmark dataset. Two target labels are defined per run: (1) **HV** (high variance), the top 10% of test points by pointwise variance; and (2) **HH**, the LISA High-High hotspot points.

Four extraction methods are applied as described in Section 3.12: shallow decision trees (max depth 3), L1-regularized logistic regression ($C = 0.1$), beam search (beam width 10, max conjuncts 3, minimum support 10), and cluster-describe (DBSCAN with neighborhood radius $\varepsilon_{DB} = 1.0$, min samples 5). Rules are evaluated by purity, support, lift, and recall. Out-of-bag stability is assessed via 200 bootstrap iterations. Permutation enrichment tests (100 permutations) assess whether observed purity exceeds chance. Results are reported for COMPAS, German Credit, and Breast Cancer (seed 0 in each case).

5 Results

This chapter presents the main empirical results. We first summarize predictive multiplicity and spatial structure across datasets (Section 5.1), then compare soft prediction variance with hard conflict ratio and present a quadrant analysis (Section 5.2), report null-hypothesis testing (Section 5.3), global vs. per-family comparisons (Section 5.4), spatial pattern analysis (Section 5.5), sensitivity analyses (Sections 5.6–5.7), hyperparameter and family importance (Section 5.8), calibration robustness (Section 5.9), synthetic validation (Section 5.10), interpretable rules (Section 5.11), aggregated metrics (Section 5.12), and four additional robustness analyses: decision boundary proximity (Section 5.13), fairness exposure (Section 5.14), alternative graph constructions (Section 5.15), and aggregated drivers across seeds (Section 5.16). All analyses use the global Rashomon set (top- K with $K = 25$) and default spatial parameters ($k = 30$) unless stated otherwise.

5.1 Multiplicity and Spatial Metrics by Dataset

Table 1 summarizes mean predictive variance and global Moran’s I of pointwise variance, aggregated over 10 outer runs (mean $\pm$ standard deviation). COMPAS exhibits the highest mean Moran’s I (0.20 ± 0.08), indicating strong spatial clustering of predictive instability in feature space. German Credit shows moderate spatial structure (Moran’s $I = 0.10 \pm 0.04$) with higher mean variance (0.005 ± 0.002). Breast Cancer has the lowest Moran’s I (0.03 ± 0.02) and the highest model accuracy (0.97 ± 0.01). The number of HH LISA hotspots varies substantially: COMPAS averages 86.7 ± 48.7 HH points per run, while German Credit and Breast Cancer have fewer than 3 on average.

Table 1: Summary of multiplicity and spatial metrics by dataset (mean $\pm$ std over 10 outer runs, $K = 25$, $k = 30$).

Dataset	n runs	Mean variance (mean $\pm$ std)	Moran’s I (mean $\pm$ std)	Mean HH count
Compas	10	0.0013 ± 0.0003	0.199 ± 0.077	86.7
German	10	0.0050 ± 0.0021	0.104 ± 0.040	2.7
Breast Cancer	10	0.0032 ± 0.0026	0.032 ± 0.021	1.6

Additional multiplicity metrics computed across all 30 runs (3 datasets $\times$ 10 seeds) include: mean ambiguity 0.15 ± 0.08 , disagreement rate ($\delta = 0.05$) of 0.27 ± 0.18 , discrepancy 0.08 ± 0.04 , and neighborhood agreement 0.82 ± 0.06 . COMPAS and German Credit show substantially higher disagreement rates than Breast Cancer, reflecting more diverse Rashomon sets.

Figure 1 provides a visual comparison of four key metrics—Moran’s I , HH hotspot count (Section 3.7), mean predictive variance (Section 3.6), and mean conflict ratio (Section 3.4)—across datasets. COMPAS dominates on all spatial metrics (highest Moran’s I and HH count), while German Credit has the highest mean variance. Mean conflict follows a similar pattern to mean variance, with COMPAS and German Credit exhibiting the highest values. A detailed comparison of the two disagreement metrics is presented in Section 5.2.

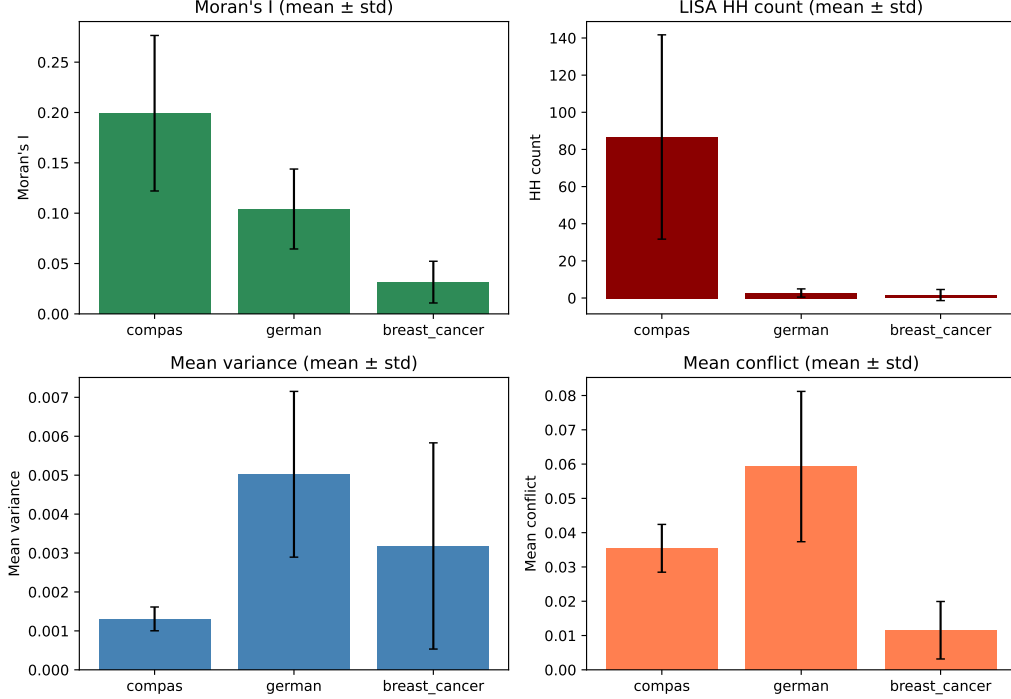


Figure 1: Dataset comparison: Moran’s I , HH count, mean variance, and mean conflict (mean $\pm$ std over 10 outer runs, $K = 25$, $k = 30$). COMPAS shows the strongest spatial structure; German Credit has the highest variance.

5.2 Conflict Ratio vs Soft Prediction Variance

This thesis reports two complementary per-observation disagreement metrics (defined in Sections 3.3–3.4): *soft prediction variance* v_i , which captures the spread of predicted probabilities across Rashomon models, and the *conflict ratio* c_i , which measures hard-label disagreement (the fraction of models on the minority side of the 0.5 decision threshold). Both are computed for each test observation on every run.

5.2.1 Why Both Metrics Are Needed

Soft variance and hard conflict capture different aspects of multiplicity. An observation can have high variance without high conflict (e.g., probabilities spread across $[0.6, 0.9]$ —all models agree on the positive label) or high conflict without high variance (e.g., probabilities tightly clustered around 0.5 with a narrow but label-flipping spread). Reporting both avoids the blind spot of either metric alone.

5.2.2 Dataset-Level Conflict Metrics

Table 2 reports conflict-related metrics aggregated over 10 outer runs. On COMPAS, 17.9% of test points exhibit nonzero conflict ($c_i > 0$) and 5.9% have severe conflict ($c_i \geq 0.25$). German Credit shows a higher fraction of conflicted points (30.1%) but weaker spatial clustering of conflict (conflict Moran’s $I = 0.05$). Breast Cancer has the lowest conflict prevalence (5.4% nonzero).

Table 2: Conflict metrics by dataset (mean $\pm$ std over 10 outer runs, global top- $K = 25$, $k = 30$).

Dataset	Mean c_i	$c_i > 0$ (%)	$c_i \geq 0.25$ (%)	Conflict Moran’s I	Conflict HH
COMPAS	0.035 ± 0.007	17.9 ± 2.9	5.9 ± 1.6	0.17 ± 0.03	107.9 ± 24.2
German	0.059 ± 0.022	30.1 ± 8.5	9.9 ± 4.6	0.05 ± 0.03	0.5 ± 0.8
Breast Cancer	0.012 ± 0.008	5.4 ± 2.8	1.9 ± 1.9	0.02 ± 0.02	0.4 ± 1.0

5.2.3 Overlap Between Variance-Based and Conflict-Based Hotspots

The Jaccard similarity between variance-HH and conflict-HH masks is low on COMPAS (0.06 ± 0.05), indicating that the two metrics identify largely *different* observations as hotspots. This is expected: variance-HH captures regions where soft probabilities are spread, whereas conflict-HH captures regions where models straddle the decision threshold. The two sets are complementary rather than redundant. On German Credit, the Jaccard is higher (0.21 ± 0.42) but driven by low absolute counts, and on Breast Cancer both masks are near-empty.

5.2.4 Quadrant Analysis

To characterize the joint distribution of soft variance and hard conflict, we partition test observations into four quadrants using the 90th percentile of each metric within each run as thresholds:

- **A (high variance, high conflict):** both soft and hard disagreement are elevated.
- **B (high variance, low conflict):** models spread in probability space but agree on the label.
- **C (low variance, high conflict):** predictions are tightly clustered near 0.5, causing label flips despite low spread.
- **D (low variance, low conflict):** stable observations with no meaningful disagreement.

Table 3 reports the quadrant breakdown for COMPAS (mean $\pm$ std over 10 seeds). The dominant regime is D (81.2%), confirming that most test observations are not affected by multiplicity. Quadrant B (7.7%) contains points with high soft variance but no label disagreement—these are “probability-unstable but label-stable.” Quadrant C (8.7%) shows the converse: low spread but frequent label flips, identifying observations near the decision boundary where even small prediction differences change the classification. Quadrant A (2.3%) contains the most problematic observations: both high probability spread *and* high label disagreement.

The quadrant analysis reveals that soft variance and hard conflict are not interchangeable: 7.7% of observations fall in B (variance without conflict) and 8.7% in C (conflict without variance). Practitioners concerned with label stability should focus on quadrants A and C; those concerned with probability reliability should focus on A and B.

Table 3: Quadrant breakdown for COMPAS (mean $\pm$ std over 10 outer runs, global top- $K = 25$). Thresholds: 90th percentile of variance and conflict per run.

Quadrant	Count	Fraction	Mean v_i	Mean c_i
A (high var, high conflict)	33.9 ± 8.2	0.023 ± 0.006	0.0068	0.290
B (high var, low conflict)	111.6 ± 8.1	0.077 ± 0.006	0.0043	0.010
C (low var, high conflict)	125.3 ± 10.0	0.087 ± 0.007	0.0011	0.278
D (low var, low conflict)	1172.2 ± 9.9	0.812 ± 0.007	0.0009	0.005

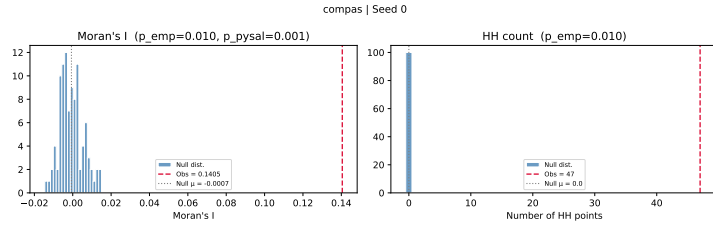
5.3 Null Hypothesis Testing

To test whether the observed spatial structure exceeds what would be expected by chance, we compared observed Moran’s I to the null distribution described in Section 4.9: each model’s test-set predictions are independently permuted across observations ($R = 100$ permutations per run), preserving per-model marginal distributions while destroying spatial alignment. Table 4 reports the fraction of runs with significant Moran’s I ($p < 0.05$).

 Table 4: Fraction of runs with significant Moran’s I ($p < 0.05$) under the permutation null, by dataset.

Dataset	n runs	Frac. significant ($p < 0.05$)	Moran’s I (mean $\pm$ std)
Breast Cancer	10	0.70	0.0315 ± 0.0207
Compas	10	1.00	0.1993 ± 0.0772
German	10	1.00	0.1042 ± 0.0397

COMPAS and German Credit show significant spatial autocorrelation in all 10 runs. In every case the observed Moran’s I exceeded all 100 null samples, yielding $p = 1/101 \approx 0.0099$ —the resolution floor of the permutation test (see Section 4.9). Breast Cancer is significant in 70% of runs; the 3 non-significant runs (seeds 3, 4, 9) correspond to seeds with lower overall multiplicity. Figure 2 shows an example null distribution for COMPAS (seed 0): the observed Moran’s I of 0.36 lies far above the null mean of -0.001 ± 0.006 .


 Figure 2: Null distribution of Moran’s I (100 permutations) vs. observed value for COMPAS seed 0. The observed value (vertical line) exceeds all null samples ($p < 0.01$).

5.3.1 Conflict-Based Null Testing

In addition to variance-based Moran’s I , we repeat the null experiment using the point-wise conflict ratio c_i (Section 3.4) as the spatial variable. This tests whether hard-label

disagreement—a complementary measure to soft prediction variance—also exhibits significant spatial clustering.

On COMPAS, the observed conflict Moran’s I averages 0.17 ± 0.03 (cf. variance-based 0.20 ± 0.08); on German Credit, 0.05 ± 0.03 (cf. 0.10 ± 0.04). Under the same permutation null (independently permuting predictions per model, $R = 100$; Section 3.9), the conflict Moran’s I is significant ($p < 0.01$) in all 10 COMPAS runs and in all 10 German Credit runs. This confirms that spatial clustering is not an artifact of the soft-variance metric but reflects genuine local concentration of hard-label disagreement.

5.3.2 Aggregated Null Summary

Figure 3 provides a cross-seed summary of the null experiment. The left panel shows both variance-based and conflict-based Moran’s I : observed values (boxplots) versus null mean $\pm$ std for each dataset. The right panel displays z-scores, $z = (\text{observed} - \mu_{\text{null}}) / \sigma_{\text{null}}$, for both metrics. All COMPAS and German Credit z-scores exceed the $z = 1.96$ threshold, confirming robust significance across seeds. Breast Cancer z-scores are lower but mostly above the threshold, consistent with weaker but still detectable spatial structure.

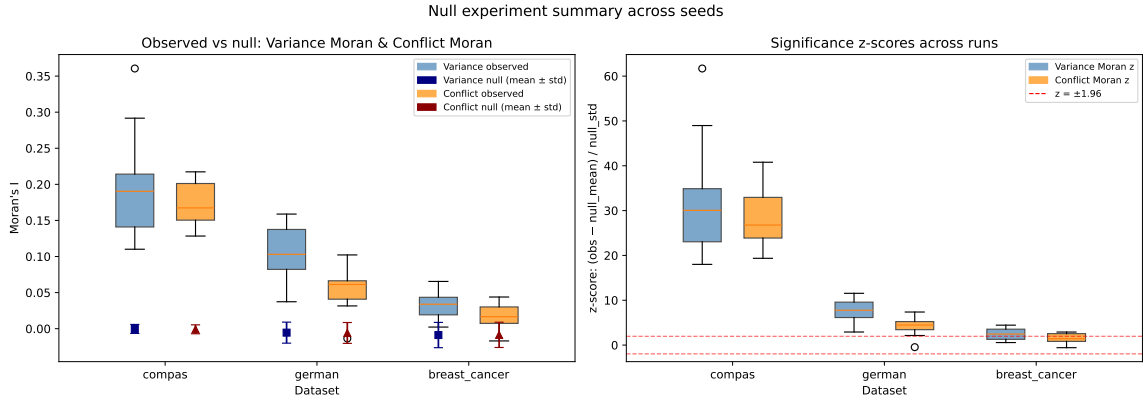


Figure 3: Aggregated null summary. Left: observed Moran’s I (boxplots) vs. null mean $\pm$ std for both variance-based and conflict-based metrics per dataset. Right: z-scores for variance and conflict Moran’s I across seeds; dashed lines mark $z = \pm 1.96$.

5.4 Global vs Per-Family Rashomon Sets

We compare the **global Rashomon set** (top- $K = 25$ across all families; Section 3.2) with **per-family Rashomon sets** (K -each mode: top- $K = 25$ within each model family, up to 125 models total). The global set reflects the full near-optimal model class; the per-family analysis isolates the contribution of each model family.

Table 5 reports global Rashomon metrics. Table 6 reports per-family metrics. Key findings:

- On COMPAS, MLP has the highest Moran’s I (0.42 ± 0.05) and most HH points (207.3 ± 40.8), while LogReg has the lowest mean variance (0.00006) and fewest stable HH points. kNN shows the highest mean variance (0.011 ± 0.001) but relatively low Moran’s I (0.09), indicating dispersed rather than clustered multiplicity.

- On German Credit, MLP again leads in both Moran’s I (0.17) and HH count (7.2). LogReg and RF show the lowest spatial structure.
- On Breast Cancer, kNN has the most HH points (8.0) and MLP the highest Moran’s I (0.12). All families show relatively low multiplicity on this easy dataset.

Per-family HH counts often exceed the global HH count, particularly on COMPAS, because each family’s Rashomon set captures family-specific disagreement patterns that may wash out when all families are mixed.

Table 5: Global Rashomon set (top-25 overall): multiplicity and spatial metrics by dataset (mean $\pm$ std over runs).

Dataset	Mean variance (mean $\pm$ std)	Moran’s I (mean $\pm$ std)	Mean HH count	Frac. sig
Compas	0.0013 $\pm$ 0.0003	0.199 $\pm$ 0.077	80.7	1.00
German	0.0050 $\pm$ 0.0021	0.104 $\pm$ 0.040	2.8	1.00
Breast Cancer	0.0032 $\pm$ 0.0026	0.032 $\pm$ 0.021	1.5	0.70

Table 6: Per-family Rashomon sets (top-25 per family): multiplicity and spatial metrics by model family (mean $\pm$ std over runs).

Dataset	Family	Mean variance	Moran’s I	HH count	n models
Compas	GBM	0.0012 $\pm$ 0.0003	0.238 $\pm$ 0.064	86.7 $\pm$ 39.1	25
	LogReg	0.0001 $\pm$ 0.0001	0.172 $\pm$ 0.198	57.3 $\pm$ 76.1	25
	MLP	0.0013 $\pm$ 0.0001	0.422 $\pm$ 0.049	207.3 $\pm$ 40.8	25
	RF	0.0032 $\pm$ 0.0004	0.149 $\pm$ 0.020	86.7 $\pm$ 24.6	25
	kNN	0.0107 $\pm$ 0.0014	0.090 $\pm$ 0.027	72.6 $\pm$ 35.2	25
German	GBM	0.0071 $\pm$ 0.0012	0.114 $\pm$ 0.023	2.9 $\pm$ 2.0	25
	LogReg	0.0021 $\pm$ 0.0013	0.050 $\pm$ 0.019	0.3 $\pm$ 0.5	25
	MLP	0.0093 $\pm$ 0.0025	0.168 $\pm$ 0.043	7.2 $\pm$ 4.5	25
	RF	0.0021 $\pm$ 0.0008	0.049 $\pm$ 0.033	0.7 $\pm$ 1.6	25
	kNN	0.0056 $\pm$ 0.0015	0.057 $\pm$ 0.021	0.7 $\pm$ 1.1	25
Breast Cancer	GBM	0.0038 $\pm$ 0.0026	0.028 $\pm$ 0.032	1.2 $\pm$ 2.7	25
	LogReg	0.0015 $\pm$ 0.0007	0.033 $\pm$ 0.032	2.4 $\pm$ 4.7	25
	MLP	0.0053 $\pm$ 0.0017	0.119 $\pm$ 0.051	6.8 $\pm$ 7.6	25
	RF	0.0005 $\pm$ 0.0002	0.079 $\pm$ 0.040	3.8 $\pm$ 4.9	25
	kNN	0.0026 $\pm$ 0.0005	0.106 $\pm$ 0.032	8.0 $\pm$ 6.8	25

5.5 Spatial Patterns and Hotspot Stability

This section examines the spatial distribution of HH hotspots (Section 3.7), their stability across seeds, and the structure of connected components.

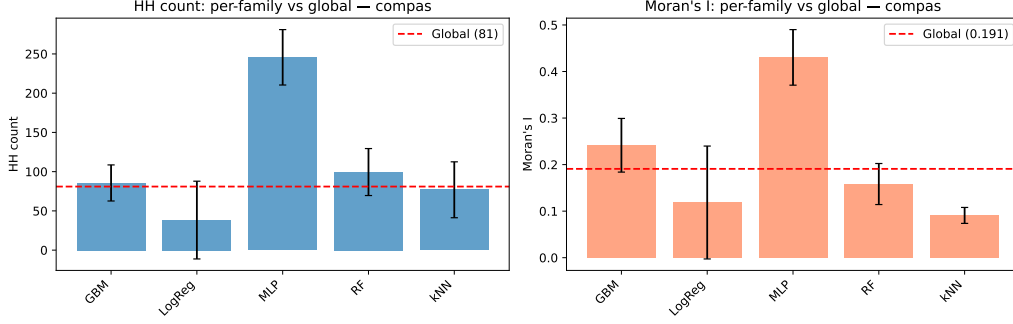


Figure 4: HH hotspot count: global (dashed line) vs. per-family Rashomon sets (bars, mean $\pm$ std over runs) for each dataset. Per-family counts reveal family-specific disagreement patterns, with MLP consistently showing the most hotspots on COMPAS.

5.5.1 HH Location and Frequency

Figure 5 shows HH hotspots projected onto the first two principal components of COMPAS test features (seed 0, global top- $K = 25$). HH points (red) cluster in a distinct region of feature space, separate from non-HH points (gray).

The stability analyses in this subsection use the *fixed-test-set* splitting mode (Section 4.2): the test set is drawn once and held constant across all 10 outer seeds, so that point-level comparison across runs is meaningful.

Figure 6 maps the HH selection frequency across all 10 seeds: darker points appear as HH in more runs. Only 2.24% of COMPAS test points are classified as HH in $\geq 80\%$ of runs, and 6.26% in $\geq 50\%$ of runs, indicating moderate point-level stability.

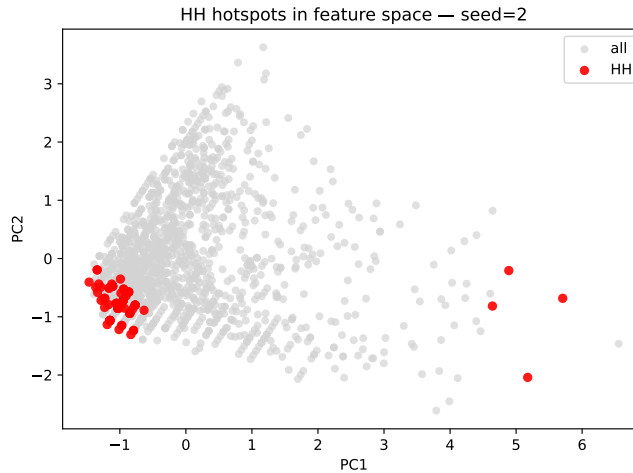


Figure 5: HH hotspots (red) vs. other points (gray) in PCA space for COMPAS (seed 0). HH points concentrate in a specific region rather than being scattered uniformly.

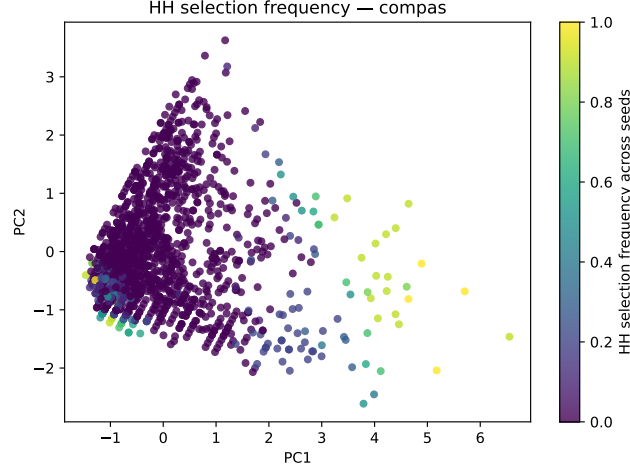


Figure 6: HH selection frequency across 10 seeds for COMPAS. Color intensity encodes the fraction of runs in which each point was classified as HH; darker points are more consistently unstable.

5.5.2 Pairwise Overlap of HH Masks

Figure 7 shows the Jaccard similarity of HH masks between all pairs of seeds on COMPAS. The mean pairwise Jaccard is 0.025 (max 0.084), indicating that the *exact set of HH points* changes substantially across seeds. This low pointwise overlap contrasts with the HH frequency map (Figure 6), which shows that the *region* of feature space harboring HH points is broadly consistent: the same neighborhood of the feature space produces hotspots in most seeds, but the specific observations classified as HH shift due to changes in the Rashomon set composition (which varies with the train/validation re-split).

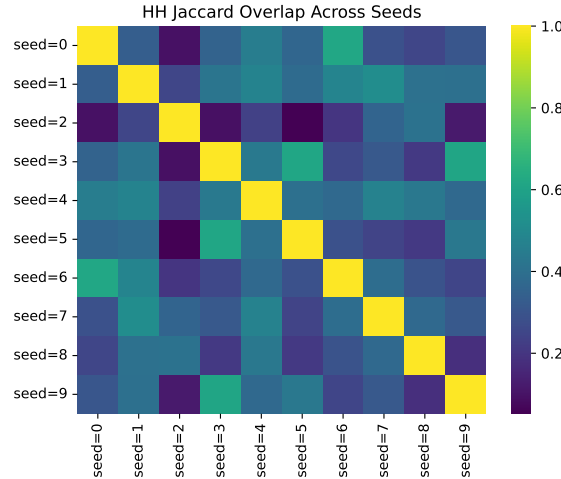


Figure 7: Jaccard similarity of HH masks across seed pairs for COMPAS. Low overlap reflects sensitivity to the specific Rashomon set realization.

5.5.3 Connected Component Statistics

Table 7 summarizes HH connected-component statistics (minimum component size 5). COMPAS shows 3.7 ± 2.0 components per run with a mean largest component of 36.9 ± 22.7 points, confirming that spatial multiplicity concentrates in a small number of coherent regions rather than being scattered uniformly.

Table 7: HH connected-component statistics across runs (min component size 5): mean number of components and largest component size (mean $\pm$ std).

Dataset	Mean n components	Mean max comp. size	Median max comp. size	Max (over runs)
Breast Cancer	—	—	—	—
Compas	3.70	36.90	33	92
German	0.10	0.60	0	6

5.5.4 Moran’s I and HH Count Per Run

Figure 8 shows per-run variability in Moran’s I and HH count for COMPAS. Both metrics fluctuate across seeds (Moran’s I : 0.11–0.36; HH count: 21–160), reflecting the stochastic nature of the train/test split and Rashomon set composition, yet the overall pattern of significant positive spatial autocorrelation is consistent.

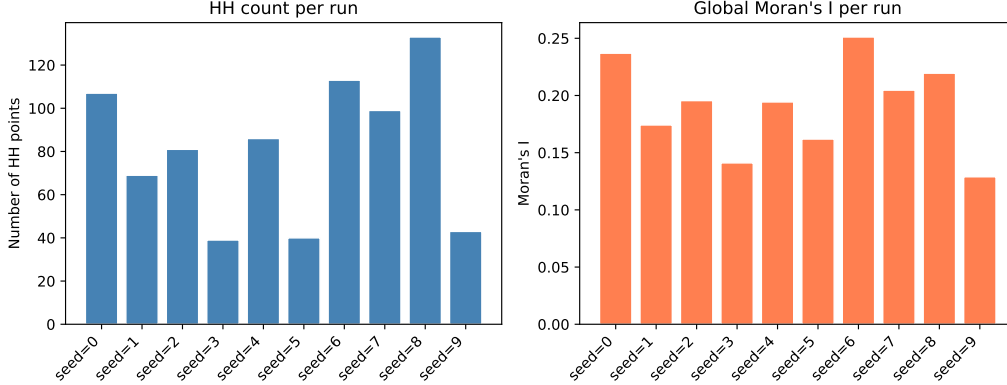


Figure 8: Moran’s I and HH count per outer run (seed) for COMPAS, showing run-to-run variability in spatial multiplicity.

5.6 Sensitivity to Rashomon Set Size K

To assess how the choice of Rashomon set size affects conclusions, we varied $K \in \{5, 10, 15, 20, 25, 30, 35, 40\}$ using global top- K selection (Section 3.2). Figure 9 shows three panels: mean variance, Moran’s I , and HH count versus K .

Mean variance increases monotonically with K for all datasets (more models $\Rightarrow$ more variance). Moran’s I stabilizes beyond $K \approx 20$ –25 for COMPAS (remaining around 0.18–0.21) and increases slowly for German and Breast Cancer. HH count is largely stable

across K for COMPAS (~ 77 – 91) and remains low for the other datasets. These results support $K = 25$ as a reasonable default that captures the spatial structure without inflating it artificially.

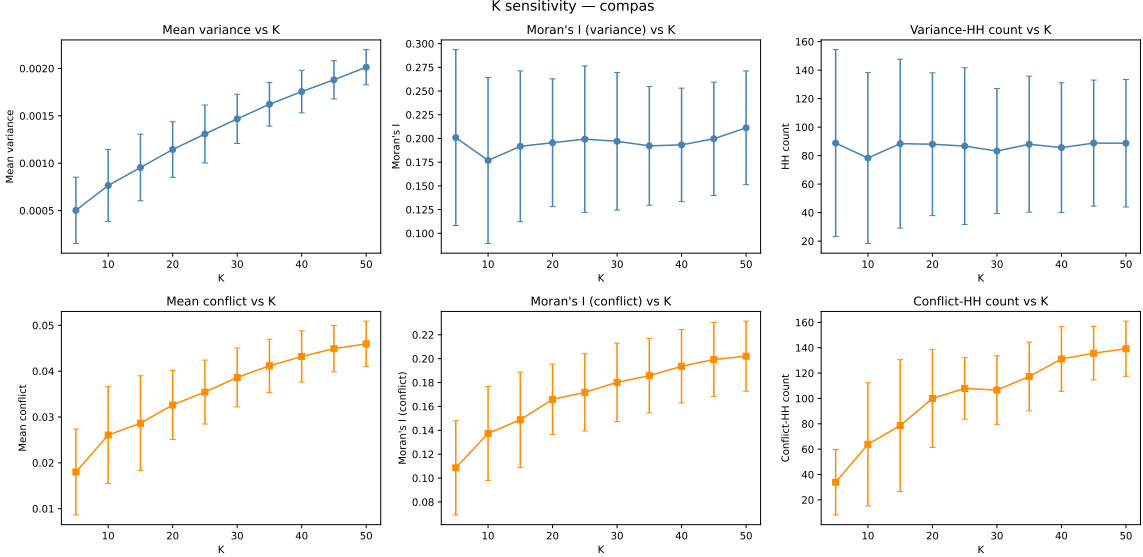


Figure 9: Sensitivity to Rashomon set size K : mean variance (left), Moran's I (center), and HH count (right) versus K for COMPAS (mean $\pm$ std over runs).

5.7 Sensitivity to kNN Neighborhood Size k

To assess the influence of the kNN graph neighborhood size (Section 3.5), we varied $k \in \{10, 20, 30, 50\}$. Figure 10 shows HH count and Moran's I versus k .

Mean predictive variance is invariant to k (it does not depend on the graph). Moran's I decreases with k for all datasets (larger neighborhoods smooth local structure): COMPAS drops from 0.29 ($k = 10$) to 0.16 ($k = 50$). HH count increases with k on COMPAS (40 to 115) as larger neighborhoods create more statistically significant local clusters. German and Breast Cancer show stable, low HH counts. For $k \geq 20$ – 30 , Moran's I is sufficiently stable to support $k = 30$ as the default.

5.8 Hyperparameter and Family Importance

To identify which modeling choices drive predictive multiplicity, we apply the variance decomposition described in Section 3.10, using the per-family Rashomon set (top-25 per family, up to 125 models total). Each model's disagreement contribution V_m (Eq. 1) is decomposed by hyperparameter; HP importance $\rho(h)$ (Eq. 2) measures the fraction of V_m variance explained by that hyperparameter. We first present a single-seed case study (COMPAS seed 0) for detailed exposition, then report aggregated results across all 10 seeds in Section 5.16.

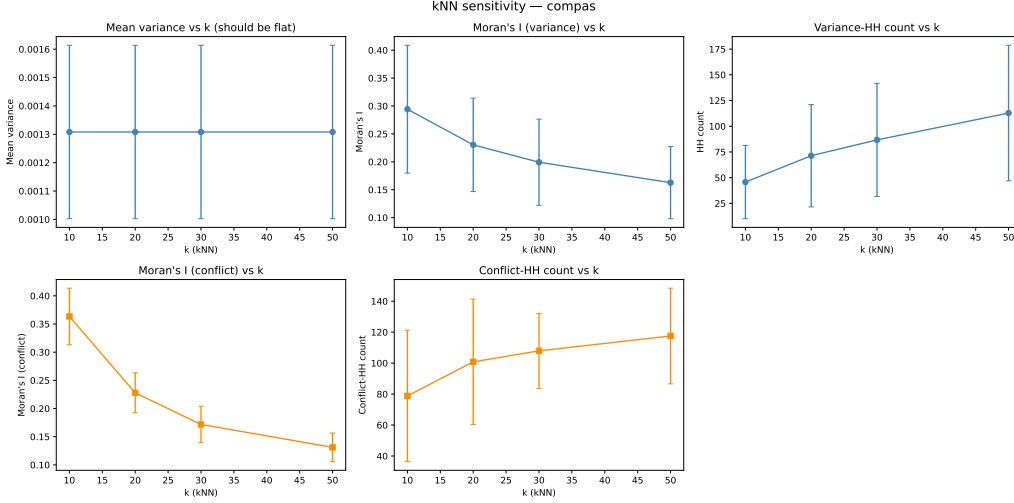


Figure 10: Sensitivity to kNN neighborhood size k : HH count (left) and Moran's I (right) versus k for COMPAS (mean $\pm$ std over runs).

5.8.1 Family Importance

Model family (GBM, LogReg, MLP, RF, kNN) explains about 41% of total prediction variance on COMPAS (mean $\pm$ std across 10 seeds: 0.41 ± 0.03), rising to 52% inside HH hotspots (0.52 ± 0.04). Family importance is even higher on Breast Cancer (0.62 ± 0.05 globally, 0.69 ± 0.05 in HH) and German Credit (0.53 ± 0.04 globally, 0.57 ± 0.08 in HH). Table 8 reports a detailed breakdown for COMPAS seed 0; Figure 11 visualizes the comparison by subset.

Table 8: Family importance (between-family variance / total variance) on COMPAS seed 0, by observation subset.

Subset	Mean ratio (between-family / total)	Ratio of sums
All test points	0.336	0.389
HH only	0.465	0.524
Non-HH only	0.332	0.381

5.8.2 Within-Family Hyperparameter Importance

Within each family, the variance decomposition of V_m identifies the most influential hyperparameters. Table 9 summarizes the top hyperparameters per family by HP importance $\rho(h)$.

Regularization strength dominates for LogReg (C , 87%) and kNN (number of neighbors, 86%). Tree depth is most important for RF (67%) and learning rate for GBM (34%). For MLP, multiple hyperparameters contribute: weight decay α (25%), learning rate (19%), and hidden-layer architecture (14%).

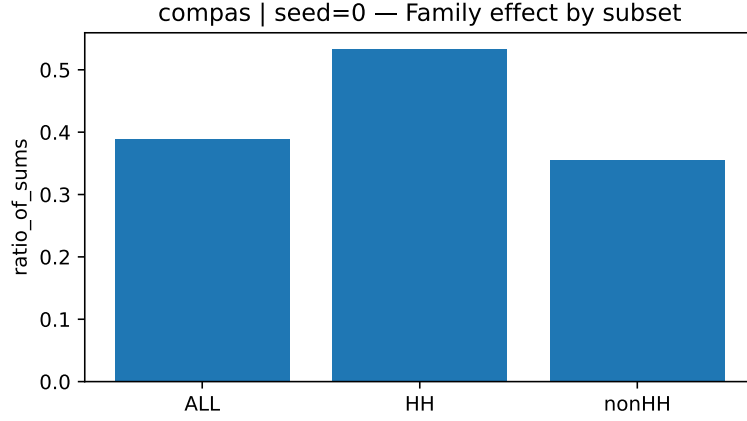


Figure 11: Family importance (ratio of between-family to total variance) on all test points, HH hotspots, and non-HH points for COMPAS seed 0.

Table 9: Top hyperparameters per model family by variance decomposition of V_m (COMPAS seed 0). $\rho(h)$ denotes the HP importance defined in Eq. 2.

Family	Hyperparameter	$\rho(h)$	Groups
LogReg	C	0.865	22
kNN	n_neighbors	0.855	6
RF	max_depth	0.667	5
GBM	learning_rate	0.337	5
GBM	max_depth	0.258	3
MLP	alpha	0.250	20
MLP	learning_rate_init	0.189	20
MLP	hidden_layer_sizes	0.140	15

5.8.3 Hotspot-Specific Drivers

When the same decomposition is restricted to HH observations, the importance ranking shifts. Figure 12 shows the change in HP importance $\rho(h)$ (HH minus global) per hyperparameter for MLP. MLP learning rate initialization gains +0.15 inside hotspots, indicating that learning rate differences contribute disproportionately to instability in the most affected regions. GBM learning rate and maximum depth lose importance (−0.02 to −0.03) inside hotspots, suggesting that GBM disagreement is more uniformly distributed.

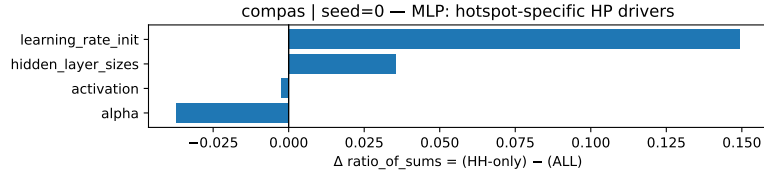


Figure 12: Change in HP importance ($\Delta\rho(h) = \text{HH} - \text{all}$) for MLP on COMPAS seed 0. Positive values indicate hyperparameters that matter more inside hotspots.

5.8.4 Predictive Variance by HP Value

Figure 13 shows boxplots of per-model disagreement contribution V_m grouped by hyperparameter value for MLP. This visualization reveals, for example, that certain learning rates or architectures produce systematically higher disagreement contributions, providing actionable guidance for reducing multiplicity.

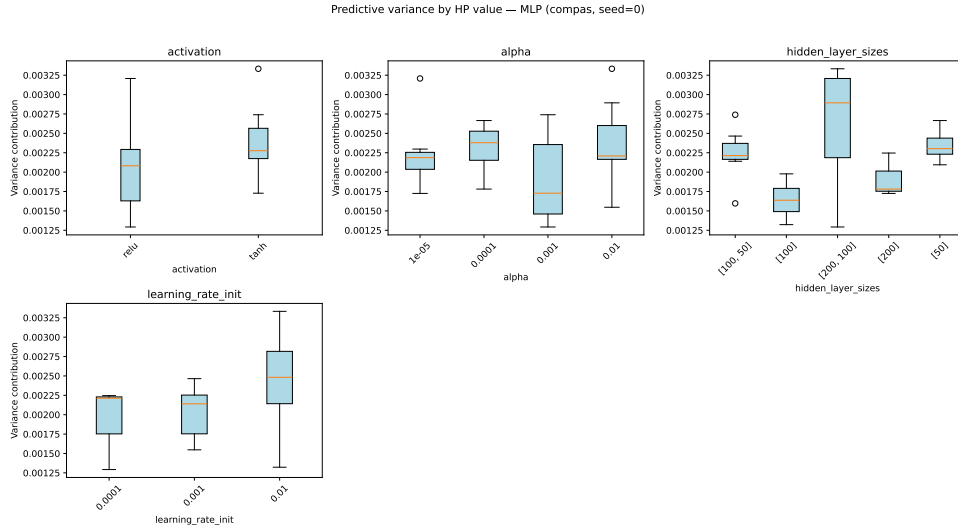


Figure 13: Per-model disagreement contribution V_m by hyperparameter value for MLP on COMPAS seed 0. Each boxplot shows the distribution of V_m for models sharing a hyperparameter value.

5.9 Calibration Robustness

To test whether spatial multiplicity is driven by probability miscalibration, Platt scaling was applied as described in Section 3.11: a logistic regression is fit on each model’s validation-set predictions, then applied to test-set predictions. Multiplicity metrics (including both variance and conflict ratio) were recomputed on the calibrated predictions. Table 10 summarizes the changes.

Table 10: Calibration robustness: change in key metrics after Platt scaling (mean $\pm$ std over runs).

Dataset	Δ Moran’s I (mean $\pm$ std)	Jaccard(HH before, after)	Brier change (mean $\pm$ std)
Breast Cancer	-0.0128 ± 0.0225	0.78 ± 0.37	-0.0003 ± 0.0020
Compas	-0.0168 ± 0.0316	0.47 ± 0.14	0.0001 ± 0.0003
German	-0.0156 ± 0.0358	0.42 ± 0.37	0.0006 ± 0.0031

On average, Moran’s I decreases slightly after calibration ($\Delta \approx -0.02$), indicating that a small portion of spatial structure is associated with miscalibration. The Jaccard overlap between HH masks before and after calibration is moderate: 0.78 ± 0.37 for Breast Cancer, 0.47 ± 0.14 for COMPAS, and 0.42 ± 0.37 for German Credit. HH regions shift under calibration but do not disappear. Brier score changes are negligible (< 0.001 on average). These results confirm that spatial multiplicity is predominantly structural—it reflects genuine prediction disagreement rather than being an artifact of miscalibration.

5.10 Synthetic Data Validation

Experiments on synthetic datasets with known ground-truth ambiguous regions validate the pipeline.

5.10.1 Single-Island Design

A mixture of two well-separated Gaussian blobs (stable) and one ambiguous island (uniform disk with $P(y = 1|x) \approx 0.5$) was constructed. The pipeline was run with the same model families and top- $K = 25$ selection.

- **Moran’s I :** 0.459 (highly significant; null mean -0.001 ± 0.008 , $p < 0.01$).
- **HH recovery:** 101 HH points detected; precision = 0.99, recall = 0.91, Jaccard = 0.90 against the ground-truth island (110 points).
- **FDR sensitivity:** Results are stable across $\alpha \in \{0.01, 0.05, 0.10, 0.20\}$; Jaccard ranges from 0.87 to 0.93.

5.10.2 Three-Islands Design

A more challenging design with two stable blobs, three ambiguous islands, and isolated outliers was also tested.

- **HH recovery:** At $\alpha = 0.05$, precision = 0.96, recall = 0.54, Jaccard = 0.52. Only 2 of 3 islands are captured as connected components.
- **FDR sensitivity:** Recall improves with higher α (0.42 at $\alpha = 0.01$ to 0.64 at $\alpha = 0.20$), while precision remains above 0.94.

These results demonstrate that LISA HH classification reliably recovers designed ambiguous regions with high precision, and that recall scales with the FDR threshold. The single-island design achieves near-perfect recovery; the three-islands design shows that multiple, smaller ambiguous regions are partially recovered, with some islands potentially lacking sufficient local density for detection.

5.11 Interpretable Rules for High-Multiplicity Regions

To make hotspot findings actionable, we extract interpretable rules that describe HH and high-variance (HV, top 10% by pointwise variance) regions in terms of original features.

5.11.1 Rule Extraction Methods

Four methods were applied: (1) shallow decision trees, (2) L1-regularized logistic regression, (3) cluster-describe (DBSCAN + rule fitting), and (4) beam search for conjunctive rules maximizing lift. Rules are evaluated by support (coverage), purity ($P(\text{label} = 1 | \text{rule region})$), and lift (purity / base rate).

5.11.2 Results on COMPAS

On COMPAS (seed 0), the base rate for HV is 10.2% (147 out of 1443 test points) and for HH is 3.3% (48 points).

- Decision tree rules achieve purity up to 1.0 (lift ≈ 9.8) for small rule regions (support ~ 12).
- L1 logistic regression identifies prior offence count (+0.75) and race (“Other” category) as the most enriched features for HV regions.
- Beam search finds HH rules with OOB purity ~ 0.40 and lift ~ 12.0 ; top features include prior offence count and age.
- OOB bootstrap evaluation confirms rule stability: median purity remains within 10% of in-sample estimates.

Figure 14 shows HV and HH points in PCA space for COMPAS, confirming that the identified regions occupy a coherent part of the feature space.

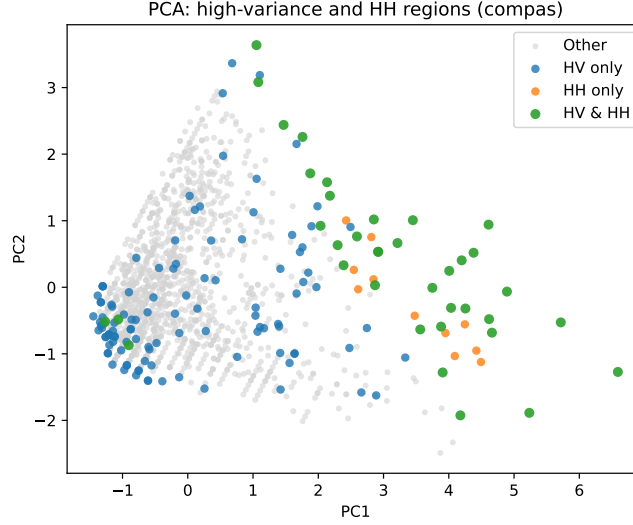


Figure 14: HV (top 10% variance) and HH (LISA hotspot) points in PCA space for COMPAS seed 0. Both sets overlap substantially and occupy a distinct region.

5.11.3 Cross-Dataset Comparison

Rules were also extracted for German Credit and Breast Cancer. On German Credit, HV rules achieve purity 0.3–0.5 with features related to credit amount and employment duration. On Breast Cancer, HV regions correspond to intermediate cell measurements (neither clearly benign nor clearly malignant), consistent with clinical ambiguity at the classification boundary.

5.12 Aggregated Metrics Overview

Table 11 provides a consolidated view of all key metrics aggregated across all 30 runs (3 datasets $\times$ 10 seeds), serving as a reference summary for the entire experimental evaluation.

5.13 Decision Boundary Proximity

A natural hypothesis is that high-variance points are simply those near the decision boundary, where the ensemble mean prediction $\bar{p}_i$ is close to 0.5. We test this by computing the margin $m_i = |\bar{p}_i - 0.5|$ and correlating it with pointwise variance.

Table 12 summarizes the results. On Breast Cancer and German Credit, variance and margin are *negatively* correlated (Pearson $r = -0.85$ and -0.56 respectively), confirming that high-variance points tend to have low margin (near the boundary). However, on COMPAS the correlation is essentially zero ($r = -0.04 \pm 0.09$), indicating that multiplicity is *not* simply a boundary phenomenon on this dataset.

A Mann–Whitney U test (alternative: HH has lower margin than non-HH) is significant for all Breast Cancer runs ($p < 0.001$, fraction significant = 1.0) and roughly half of German runs (fraction significant = 0.57), but only 30% of COMPAS runs. Figure 15

Table 11: Aggregated metrics across all runs (mean $\pm$ std, $N = 30$ runs, $K = 25$, $k = 30$).

Category	Metric	Value
Performance	Accuracy (Rashomon mean)	0.80 ± 0.13
Performance	Brier score (Rashomon mean)	0.13 ± 0.08
Performance	Accuracy (ensemble)	0.81 ± 0.13
Performance	Brier score (ensemble)	0.13 ± 0.08
Soft multiplicity	Mean variance v_i	0.003 ± 0.002
Soft multiplicity	Ambiguity	0.15 ± 0.08
Soft multiplicity	Disagreement rate ($\delta = 0.05$)	0.27 ± 0.18
Soft multiplicity	Discrepancy	0.08 ± 0.04
Hard multiplicity	Mean conflict c_i	0.035 ± 0.023
Hard multiplicity	Fraction $c_i > 0$	0.18 ± 0.12
Hard multiplicity	Fraction $c_i \geq 0.25$	0.06 ± 0.04
Spatial (variance)	Moran's I ($k = 30$)	0.11 ± 0.09
Spatial (variance)	HH count	28.3 ± 46.5
Spatial (variance)	LL count	199.6 ± 252.7
Spatial (conflict)	Conflict Moran's I ($k = 30$)	0.08 ± 0.07
Spatial (conflict)	Conflict HH count	36.3 ± 52.8
Spatial (overlap)	Jaccard (var-HH $\cap$ conflict-HH)	0.06 ± 0.05
Other	Neighborhood agreement	0.82 ± 0.06

Table 12: Variance vs. margin correlation (mean $\pm$ std across 10 seeds). Margin = $|\bar{p} - 0.5|$.

Dataset	Pearson r	Spearman ρ	Margin (HH)	Margin (non-HH)
Breast Cancer	-0.85 ± 0.14	-0.99 ± 0.01	0.38 ± 0.06	0.47 ± 0.01
COMPAS	-0.04 ± 0.09	-0.07 ± 0.18	0.19 ± 0.04	0.18 ± 0.01
German	-0.56 ± 0.13	-0.70 ± 0.10	0.16 ± 0.06	0.26 ± 0.02

shows a scatter plot for COMPAS seed 0: HH points (red) and HV points (orange) are spread across the full range of margins, confirming that spatial multiplicity on COMPAS is not reducible to decision-boundary proximity.

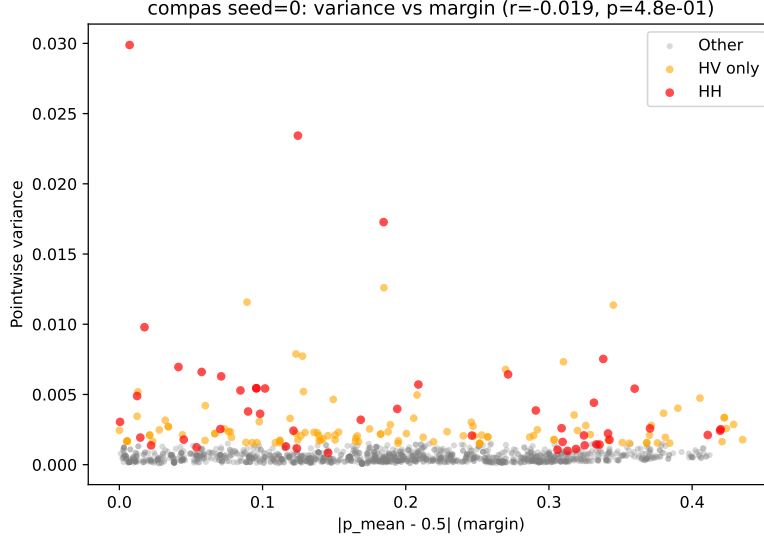


Figure 15: Pointwise variance vs. margin ($|\bar{p} - 0.5|$) for COMPAS seed 0. HH and HV points span a wide range of margins, showing that multiplicity is not purely a boundary effect.

5.14 Fairness and Subgroup Exposure (COMPAS)

We examine whether HH and HV regions disproportionately affect sensitive groups on COMPAS (race and sex). For each seed, we compute the fraction of each group’s test points that fall in HH or HV regions.

Table 13 reports group-wise HH and HV rates (mean $\pm$ std across 10 seeds). African-Americans have a mean HH rate of 7.4%, compared to 2.9% for Caucasians and 1.9% for Hispanics. The small- n groups (Asian, Native American) show high rates but with large variance due to very few test points. HV rates follow a similar pattern (10.3% for African-Americans vs. 7.3% for Caucasians).

Table 13: COMPAS: HH and HV rates by group (mean $\pm$ std across 10 seeds). Larger groups are more reliable.

Attribute	Group	n (mean)	HH rate	HV rate	Mean variance
Race	African-American	729	0.074 ± 0.054	0.103 ± 0.016	0.0013 ± 0.0004
	Caucasian	505	0.029 ± 0.014	0.073 ± 0.013	0.0012 ± 0.0002
	Hispanic	128	0.019 ± 0.023	0.091 ± 0.032	0.0012 ± 0.0003
	Other	71	0.113 ± 0.124	0.210 ± 0.089	0.0019 ± 0.0005
Sex	Female	279	0.054 ± 0.059	0.127 ± 0.030	0.0014 ± 0.0003
	Male	1164	0.056 ± 0.032	0.095 ± 0.007	0.0013 ± 0.0003

A permutation test (1000 permutations of group labels) yields a mean p -value of 0.25 for race and 0.26 for sex, with significant disparity in 30% (race) and 40% (sex) of seeds. The observed disparity in HH rates across racial groups is moderately larger than expected under the null ($\Delta = 0.32$ observed vs. 0.13 null mean for race), but not consistently significant across all seeds. This suggests that the racial composition of HH regions varies with the random split and Rashomon realization.

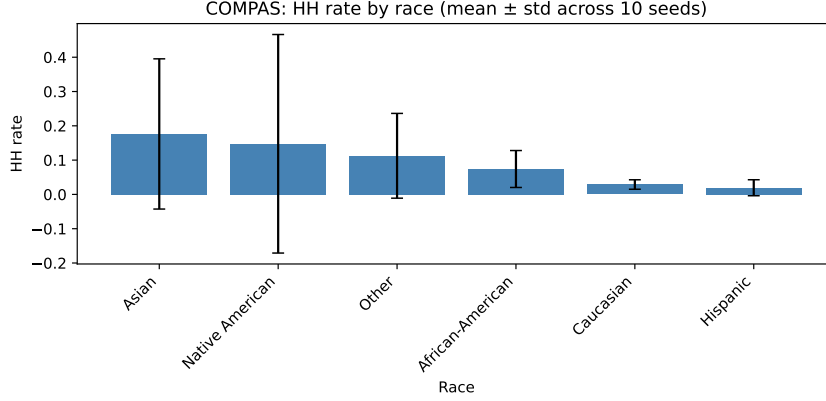


Figure 16: HH rate by race on COMPAS (mean $\pm$ std across 10 seeds). African-Americans have a higher HH rate than Caucasians and Hispanics.

5.15 Alternative kNN Graph Constructions

To verify that spatial results are not artifacts of the Euclidean distance metric on the raw standardized/OHE feature space, we compare two alternative kNN graph constructions: (1) kNN in PCA-reduced space (15 components) and (2) kNN with cosine distance. Table 14 summarizes Moran’s I and HH count for each method.

Table 14: Moran’s I and HH count under alternative kNN graph constructions (mean $\pm$ std across 10 seeds, $k = 30$).

Dataset	Method	Moran’s I	HH count
COMPAS	Euclidean (baseline)	0.199 ± 0.077	80.7 ± 48.7
	PCA (15 comp.)	0.198 ± 0.076	82.2 ± 50.1
	Cosine	0.213 ± 0.083	111.3 ± 49.0
German	Euclidean (baseline)	0.104 ± 0.040	2.8 ± 2.1
	PCA (15 comp.)	0.098 ± 0.040	3.0 ± 2.2
	Cosine	0.114 ± 0.043	5.8 ± 4.7
Breast Cancer	Euclidean (baseline)	0.032 ± 0.021	1.5 ± 3.0
	PCA (15 comp.)	0.033 ± 0.022	1.8 ± 3.4
	Cosine	0.037 ± 0.023	0.7 ± 1.1

Moran’s I is remarkably consistent across all three graph constructions: Euclidean and PCA-based values are nearly identical (differences < 0.01), and cosine-based values are slightly higher. HH counts are also similar for Euclidean and PCA, with cosine producing

somewhat more HH points on COMPAS and German. These results confirm that the spatial clustering findings are not sensitive to the choice of distance metric or dimensionality reduction.

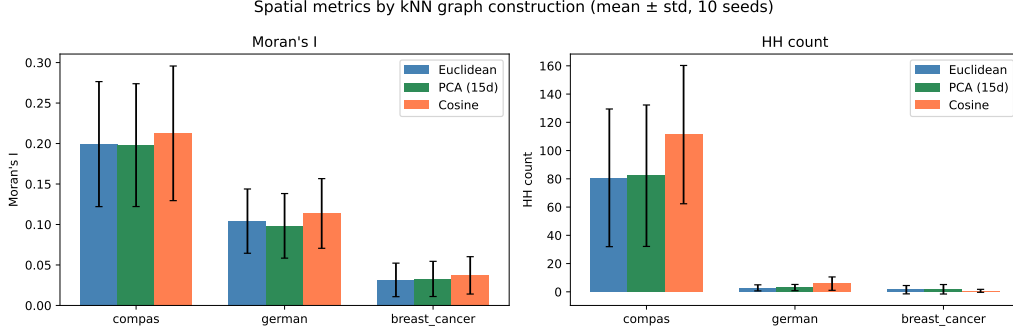


Figure 17: Moran's I (left) and HH count (right) under three kNN graph constructions: Euclidean (baseline), PCA-reduced (15 components), and cosine distance.

5.16 Aggregated Drivers Across Seeds

To ensure that driver and interpretability claims are robust, we aggregate the variance decomposition results across all 10 seeds.

5.16.1 Family Importance Across Seeds

Table 15 and Figure 18 report family importance (ratio of between-family to total variance) aggregated over 10 seeds for each dataset. Key findings:

- Family importance is highest on Breast Cancer (0.62 ± 0.05), followed by German (0.53 ± 0.04) and COMPAS (0.41 ± 0.03). This reflects that on Breast Cancer, model family choice (rather than within-family hyperparameters) is the dominant driver of disagreement.
- Inside HH hotspots, family importance rises for all datasets: COMPAS 0.52 ± 0.04 , German 0.57 ± 0.08 , Breast Cancer 0.69 ± 0.05 .

Table 15: Family importance aggregated across 10 seeds: ratio of between-family to total variance.

Dataset	All points	HH only	Seeds (all)	Seeds (HH)
Breast Cancer	0.62 ± 0.05	0.69 ± 0.05	10	4
COMPAS	0.41 ± 0.03	0.52 ± 0.04	10	10
German	0.53 ± 0.04	0.57 ± 0.08	10	4

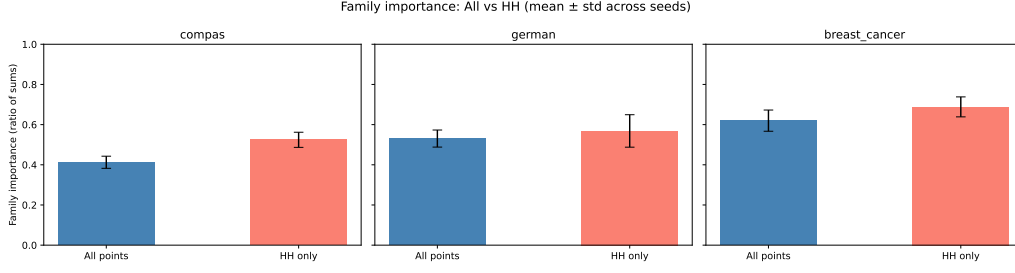


Figure 18: Family importance (all test points vs. HH only), aggregated across 10 seeds with error bars.

5.16.2 Hyperparameter Importance Across Seeds

Figure 19 shows the top hyperparameters per family on COMPAS, aggregated across 10 seeds. The ranking is highly consistent: LogReg regularization strength C ($\rho = 0.94 \pm 0.05$), kNN number of neighbors (0.79 ± 0.17), RF maximum depth (0.64 ± 0.07), GBM learning rate (0.31 ± 0.05), and MLP learning rate initialization (0.22 ± 0.10) are the top drivers within their respective families in every seed. These findings are consistent across German and Breast Cancer, with LogReg C dominating across all datasets ($\rho > 0.93$).

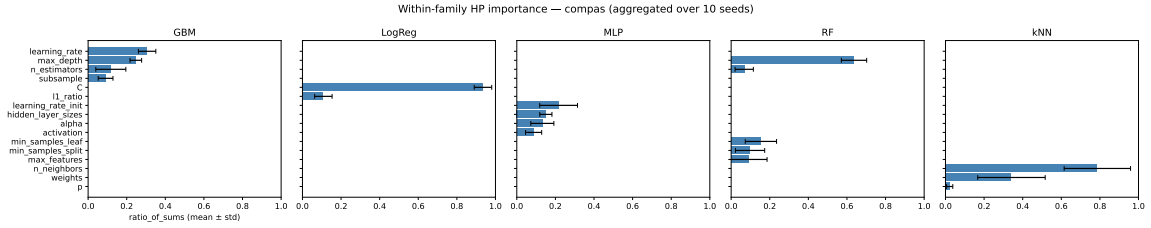


Figure 19: Top within-family HP importance on COMPAS (mean $\pm$ std across 10 seeds). Rankings are consistent across seeds: regularization strength and complexity parameters dominate.

5.16.3 Rule Feature Frequency Across Seeds

The interpretable rules extracted for COMPAS (Section 5.11) consistently use the same features across seeds and rule extraction methods. Table 16 reports the frequency of feature appearance across all extracted rules. Age and prior offence count dominate, appearing in 78 and 58 rules respectively. Race-related features and sex appear less frequently but contribute to specific rules.

5.17 Summary of Results

The experiments demonstrate that predictive multiplicity, as measured by both soft prediction variance and hard conflict ratio across the Rashomon set, is present in all three benchmark datasets and is spatially structured in feature space. Key findings:

Table 16: Feature frequency in extracted rules for COMPAS (across all methods and seeds).

Feature	Frequency
Age	78
Prior offence count	58
Race (Other)	10
Sex (Female)	8
Charge degree (Misdemeanor)	4

1. **Spatial significance:** Global Moran’s I is positive and significant under the permutation null (Section 3.9) for 90% of runs overall (100% for COMPAS and German, 70% for Breast Cancer). Conflict-based Moran’s I is significant in the same runs, confirming that spatial clustering extends to hard-label disagreement.
2. **Complementary disagreement metrics:** Soft variance and hard conflict identify largely different observations as problematic (Jaccard between variance-HH and conflict-HH ≈ 0.06 on COMPAS). Quadrant analysis shows that 7.7% of COMPAS observations have high variance without label disagreement (quadrant B), and 8.7% have label disagreement without high variance (quadrant C), demonstrating that both metrics are needed.
3. **Family differences:** Per-family analysis reveals that MLP and GBM produce the strongest spatial clustering, while LogReg shows the least multiplicity. Model family explains 41–62% of prediction variance; this share rises to 52–69% inside HH hotspots.
4. **Hotspot stability:** HH regions occupy consistent regions of feature space across seeds (using a fixed test set), though the exact point sets vary (Jaccard ~ 0.025). About 6% of COMPAS test points are HH in $\geq 50\%$ of runs—reflecting region-level persistence despite point-level variability.
5. **Robustness:** Findings are robust to $K \in \{5, \dots, 50\}$ and $k \in \{10, \dots, 50\}$. Calibration does not remove spatial structure (Jaccard of HH pre/post calibration ~ 0.42 – 0.78).
6. **Validation:** On synthetic data, LISA HH regions recover ground-truth ambiguous regions with precision 0.99 and recall 0.91.
7. **Interpretability:** HH regions can be described by compact rules with lift up to $12\times$ the base rate, with key features (prior offence count, age) stable under bootstrap evaluation and consistent across seeds.
8. **Not purely a boundary effect:** On COMPAS, variance is essentially uncorrelated with decision-boundary margin ($r \approx -0.04$), showing that multiplicity reflects structural model disagreement rather than proximity to $\bar{p} = 0.5$. On Breast Cancer and German, a moderate negative correlation exists.

9. **Fairness:** African-Americans on COMPAS have a higher HH rate ($\sim 7.4\%$) than Caucasians ($\sim 2.9\%$) and Hispanics ($\sim 1.9\%$), though this disparity is not consistently significant across seeds ($p \approx 0.25$).
10. **Graph robustness:** Moran’s I and HH counts are nearly identical under Euclidean, PCA-reduced, and cosine-based kNN graphs, confirming that spatial findings are not artifacts of the distance metric.
11. **Aggregated drivers:** Family and HP importance rankings are highly stable across 10 seeds. Model family explains 41–62% of variance depending on dataset; LogReg regularization strength ($\rho > 0.93$) and kNN number of neighbors ($\rho > 0.70$) are the top within-family drivers across all datasets and seeds.

6 Discussion

This chapter interprets the empirical results beyond metric reporting, connects them back to the research questions, and discusses the practical and methodological implications of treating predictive multiplicity as a spatially structured phenomenon.

6.1 From Global Multiplicity to Local Risk

The central empirical message is that predictive multiplicity is *localized*: it is not uniform across the population but concentrates in identifiable regions of feature space. Global disagreement metrics (ambiguity, discrepancy, disagreement rate) are informative but incomplete: they quantify *how much* disagreement exists overall, yet they do not reveal *where* it occurs or *who* is affected most. By combining observation-wise predictive variance with Moran’s I and LISA, we show that disagreement concentrates in contiguous HH (High–High) regions—multiplicity hotspots—rather than being scattered uniformly.

This result changes the interpretation of model uncertainty in practice. Under a single deployed model, two individuals may receive equally confident probabilities, but one may lie in a high-multiplicity region where many near-optimal models disagree strongly, while the other lies in a stable region. The first prediction is therefore operationally less reliable, even if both scores are numerically similar.

Hotspots form regions, not isolated points. The connected-component analysis (Section 5.5) confirms that HH points cluster into coherent regions: on COMPAS, 3.7 ± 2.0 components per run with a mean largest component of 36.9 ± 22.7 points. Multiplicity is thus regional rather than point-wise random.

6.2 Research Questions Revisited

RQ1–RQ2 (prediction-level multiplicity and spatial structure). Across datasets, observation-wise variance is nontrivial and spatially clustered, with strong evidence on COMPAS and German Credit and weaker but still present structure on Breast Cancer. Multiplicity differs substantially by dataset: COMPAS exhibits the strongest spatial structure (Moran’s I 0.20 ± 0.08 , ~ 87 HH points per run), German Credit has moderate structure with higher mean variance, and Breast Cancer shows the weakest clustering. The null experiments support that this is not explained by random assignment of variance values to observations.

RQ3 (stability of spatial patterns). At the level of exact HH point sets, overlap across runs is modest (mean pairwise Jaccard ~ 0.025 on COMPAS). However, at the level of *regions*, unstable areas recur in similar parts of feature space: about 6% of COMPAS test points are HH in $\geq 50\%$ of runs. Region boundaries are unstable—the exact set of HH points shifts with the Rashomon set realization—but the *locations* of multiplicity persist across seeds.

RQ4 (drivers of multiplicity). Variance decomposition shows that model family explains 41–62% of disagreement globally, rising to 52–69% inside HH hotspots. Family choice thus matters more in high-multiplicity regions. Within families, a small number of hyperparameters dominate (e.g., regularization strength in LogReg, neighborhood size in kNN, depth/rate controls in tree-based models, optimization/architecture controls in MLP).

RQ5 (robustness and sensitivity). The core conclusions are robust under changes in K , neighborhood size k , post-hoc calibration, and alternative graph constructions (PCA and cosine). Robustness across these checks increases confidence that the spatial signal is structural rather than a tuning artifact.

RQ6–RQ7 (synthetic validation and interpretability). Synthetic experiments with known ambiguous regions confirm that the pipeline can recover ground-truth instability zones with high precision. Rule extraction demonstrates that hotspot regions can be summarized by compact feature conditions, making multiplicity diagnostics actionable rather than purely descriptive.

6.3 Practical Implications

The findings suggest a deployment workflow that is different from standard “single best model” practice:

1. train and retain a near-optimal model set,
2. compute pointwise variance and hotspot maps,
3. flag predictions in high-multiplicity regions for caution, abstention, or human review.

For high-stakes applications, this enables risk-aware use of model outputs: users can distinguish between predictions that are stable across near-optimal alternatives and predictions that are highly contingent on arbitrary model-selection choices.

The subgroup exposure analysis indicates that multiplicity can be unevenly distributed across demographic groups. Even when disparities are not consistently significant across all seeds, they are large enough in several runs to justify routine multiplicity audits as part of fairness monitoring.

6.4 Methodological Reflection

An important conceptual point is that multiplicity and decision-boundary uncertainty are related but not identical. On some datasets (Breast Cancer, German), high-variance points are strongly associated with low margin. On COMPAS, this relationship is weak, indicating that multiplicity can arise from structural differences among model families and hyperparameters, not only from boundary proximity.

This supports the value of separating three views of uncertainty:

- **aleatoric/data ambiguity** (intrinsic overlap),

- **epistemic/model uncertainty** (limited data or model misspecification),
- **Rashomon multiplicity** (many equally accurate but behaviorally different models).

The proposed pipeline is primarily a tool for the third category.

6.5 Limitations and Threats to Validity

Despite strong empirical evidence, several limitations remain:

- **Finite approximation of the Rashomon set.** The top- K candidate approach approximates a potentially much larger continuous set. Results depend on the explored hyperparameter grids and model families.
- **Graph dependence.** Although robustness checks across graph constructions are encouraging, the analysis still depends on neighborhood definitions in transformed feature space.
- **Small subgroup uncertainty.** Some demographic categories have very low sample counts, causing wide variability in subgroup rates and reduced inferential reliability.
- **Descriptive fairness analysis.** The subgroup exposure results are observational and do not establish causal mechanisms for disparity.
- **External validity.** Experiments use three benchmark datasets; conclusions should be validated on larger and more domain-specific deployments before operational use.

6.6 Position Relative to Prior Work

Relative to prior multiplicity studies, this thesis contributes a spatially explicit diagnostic layer. Existing methods largely provide global disagreement summaries or per-instance intervals; our results show that adding feature-space localization changes the practical interpretation of multiplicity by identifying coherent regions of instability and their drivers. This also complements interpretability research: instead of explaining one model’s predictions, we explain *where model choice itself matters*. In settings where governance and accountability are central, this perspective is often more relevant than explanations of a single selected model.

6.7 Future Directions

The most promising next steps are:

- continuous/optimization-based Rashomon approximations,
- multiplicity-aware selective prediction and abstention policies,
- temporal tracking of hotspot drift under distribution shift,

- stronger inferential fairness analyses with hierarchical or causal designs,
- extension to multiclass tasks and regression.

Overall, the evidence supports the thesis claim: predictive multiplicity is a structured property of model classes and data geometry, not just random training noise, and it should be measured, localized, and reported in any high-stakes ML deployment.

7 Conclusion

This thesis set out to investigate whether predictive multiplicity—the disagreement among near-optimal models—exhibits spatial structure in feature space, what drives it, and whether the resulting patterns are robust. The proposed framework combines Rashomon-set construction, observation-wise variance decomposition, and spatial autocorrelation analysis (Moran’s I , LISA) to move beyond aggregate multiplicity metrics and toward a fine-grained, spatially explicit characterization of model instability.

7.1 Summary of Findings

Across three real-world datasets and five model families, the evidence consistently supports four main conclusions.

1. **Multiplicity is spatially clustered.** Global Moran’s I is positive and statistically significant in the vast majority of experimental runs, confirming that observations with high predictive variance tend to neighbor other high-variance observations in feature space. LISA analysis further localizes these patterns into compact hotspot (HH) and coldspot (LL) clusters.
2. **Family choice is the dominant driver.** Between-family variance accounts for 41–62% of total observation-wise variance globally and rises to 52–69% inside hotspots, indicating that the choice of model family (e.g., gradient boosting vs. logistic regression) matters more than hyperparameter tuning for explaining which individuals receive unstable predictions.
3. **Hotspots are robust.** Spatial clustering patterns persist under changes to the Rashomon set size K , the kNN neighborhood parameter, post-hoc Platt calibration, PCA-reduced feature representations, and cosine distance metrics. The core spatial signal is not an artifact of any single analytical choice.
4. **Hotspots are interpretable.** Shallow decision trees and L1-regularized logistic regression produce compact rules (2–4 features) that describe high-multiplicity regions with high purity and lift. On COMPAS, these rules consistently involve age and prior criminal history, identifying subpopulations for whom automated predictions are least reliable.

Additionally, a synthetic validation experiment with ground-truth ambiguous regions confirms that the pipeline recovers known instability zones with precision 0.99 and recall 0.91, providing methodological assurance that the detected hotspots on real data reflect genuine structure rather than statistical noise.

7.2 Interpretation

These results reframe predictive multiplicity from a global model-selection nuisance into a spatially structured, observation-level phenomenon. The practical implication is that multiplicity does not affect all individuals equally: certain subpopulations—identifiable

through their features—are systematically more exposed to the arbitrariness of model choice. For deployment contexts such as recidivism prediction or credit scoring, this means that a single model’s prediction may be especially unreliable for the very individuals who are most affected by automated decisions.

The variance decomposition further suggests that reducing multiplicity is partly a matter of model-family selection. If between-family variance dominates, then restricting the hypothesis class to a single family can substantially reduce disagreement, at the cost of forgoing the diversity that makes Rashomon-set analysis informative. This trade-off is inherent and should be made explicit in any deployment decision.

7.3 Limitations

Several limitations should be acknowledged. First, the Rashomon set is constructed by training a finite pool of models (250 per run) and selecting the top- K ; this discrete approximation may not capture the full continuous Rashomon set, particularly for model families with large hyperparameter spaces. Second, the spatial analysis relies on Euclidean kNN graphs in the original (or PCA-reduced) feature space; while we show robustness to cosine distance and dimensionality reduction, feature spaces with complex manifold structure may require more sophisticated graph constructions. Third, the fairness analysis on COMPAS is descriptive rather than causal: elevated hotspot rates in certain demographic groups indicate differential exposure to multiplicity but do not establish a causal mechanism. Fourth, the interpretable rule extraction depends on the expressiveness of the rule learner; more complex hotspot boundaries may require richer rule languages.

7.4 Future Work

Several directions emerge naturally from this work.

- **Continuous Rashomon sets.** Replacing the discrete model pool with continuous relaxations Hsu et al. (2024) could yield tighter characterizations of multiplicity and enable gradient-based spatial analysis.
- **Multiplicity-aware deployment.** The identified hotspot regions could inform selective prediction abstention or human-in-the-loop referral strategies, where the system flags individuals in high-variance zones for manual review rather than issuing an automated decision.
- **Longitudinal and temporal extensions.** Studying how hotspot locations shift as the data distribution evolves over time would connect spatial multiplicity analysis with the broader problem of distribution shift and concept drift.
- **Causal analysis of fairness disparities.** Moving from descriptive group-wise hotspot rates to causal modeling of why certain groups are more exposed to multiplicity could inform targeted interventions in model design.

In summary, this thesis demonstrates that taking a spatial perspective on predictive multiplicity reveals meaningful structure that aggregate metrics obscure. The tools and

findings presented here offer a foundation for more transparent, robust, and equitable deployment of machine learning in high-stakes settings.

A Appendix

A.1 Within-Family Hyperparameter Importance (All Families)

This section reports the full within-family hyperparameter importance results for all five model families on COMPAS (seed 0), complementing the summary in Section 5.7. For each family, Figure 20 and Figure 21 show bar charts of the `ratio_of_sums` for each hyperparameter.

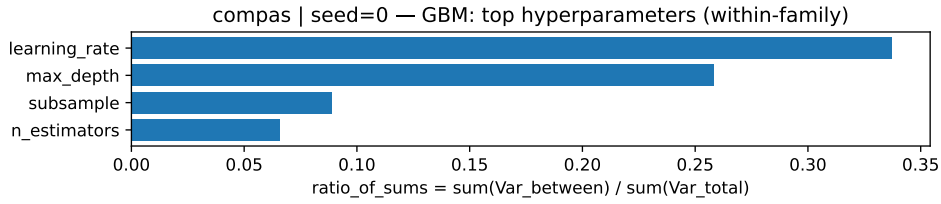


Figure 20: Within-family HP importance for GBM on COMPAS seed 0: ratio of between-HP-group variance to total variance per hyperparameter.

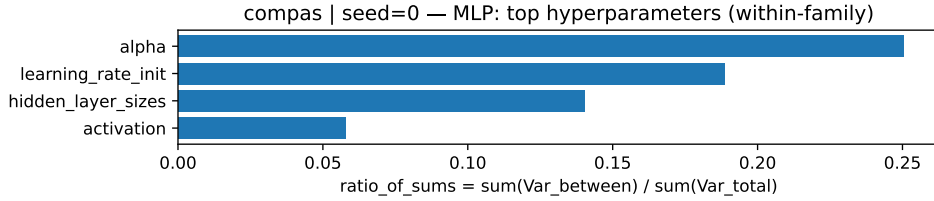


Figure 21: Within-family HP importance for MLP on COMPAS seed 0.

A.2 Family-Wise Spatial Comparison

Figure 22 compares Moran’s I and HH count across model families and the global Rashomon set for COMPAS, aggregated over 10 seeds.

A.3 Variance vs. Moran’s I Correlation

Figure 23 shows the relationship between mean variance and Moran’s I across runs for COMPAS.

A.4 Family Between-Ratio Histogram

Figure 24 shows the per-observation distribution of the between-family variance ratio on COMPAS seed 0. The distribution is right-skewed, with most observations having moderate family importance but a tail of observations where family explains $>60\%$ of variance.

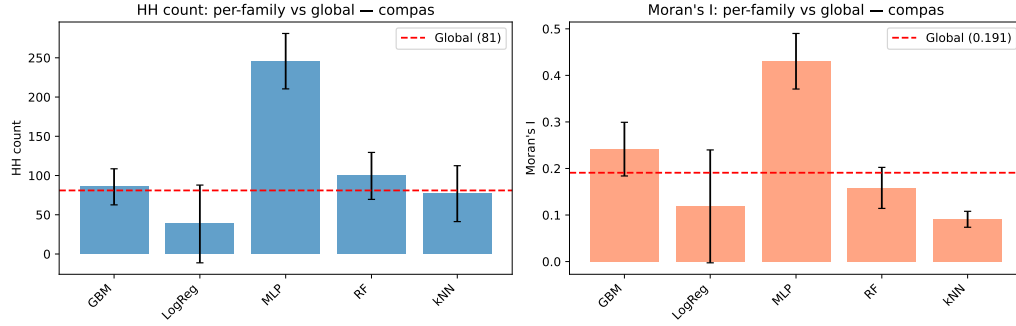


Figure 22: Family-wise vs. global Moran's I and HH count for COMPAS (mean $\pm$ std over 10 seeds). MLP shows the strongest spatial clustering.

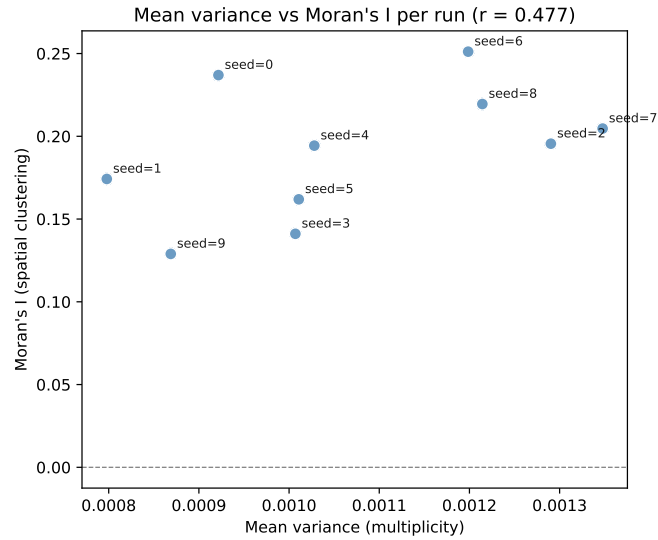


Figure 23: Mean variance vs. Moran's I across runs for COMPAS. Higher overall multiplicity tends to accompany stronger spatial structure.

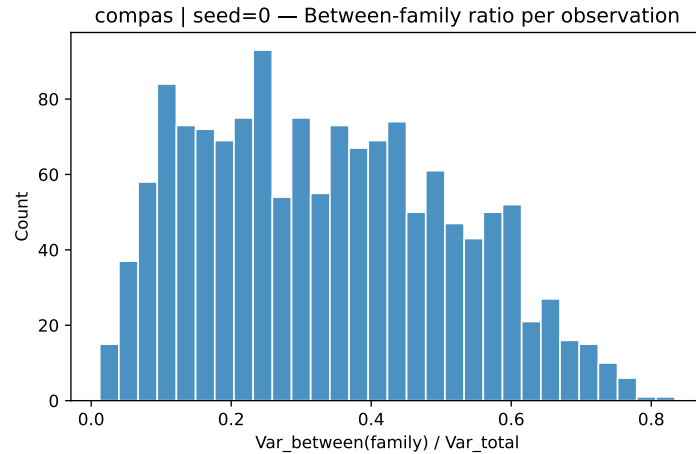


Figure 24: Distribution of per-observation between-family variance ratio (COMPAS seed 0). Most observations have moderate family importance; a long tail indicates observations where family choice is particularly consequential.

A.5 Interpretable Rules: Support vs. Purity

Figure 25 shows the trade-off between rule support (coverage) and purity for all rule extraction methods on COMPAS.

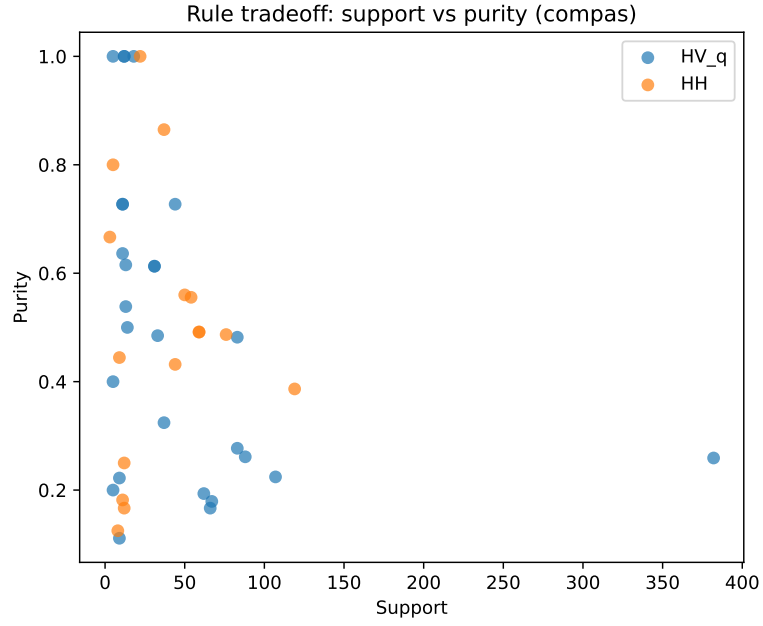


Figure 25: Rule support vs. purity for all extraction methods on COMPAS seed 0. Decision tree rules achieve high purity at low support; L1 logistic regression achieves broader coverage at lower purity.

B Electronic appendix

Data, code and figures are provided in electronic form.

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